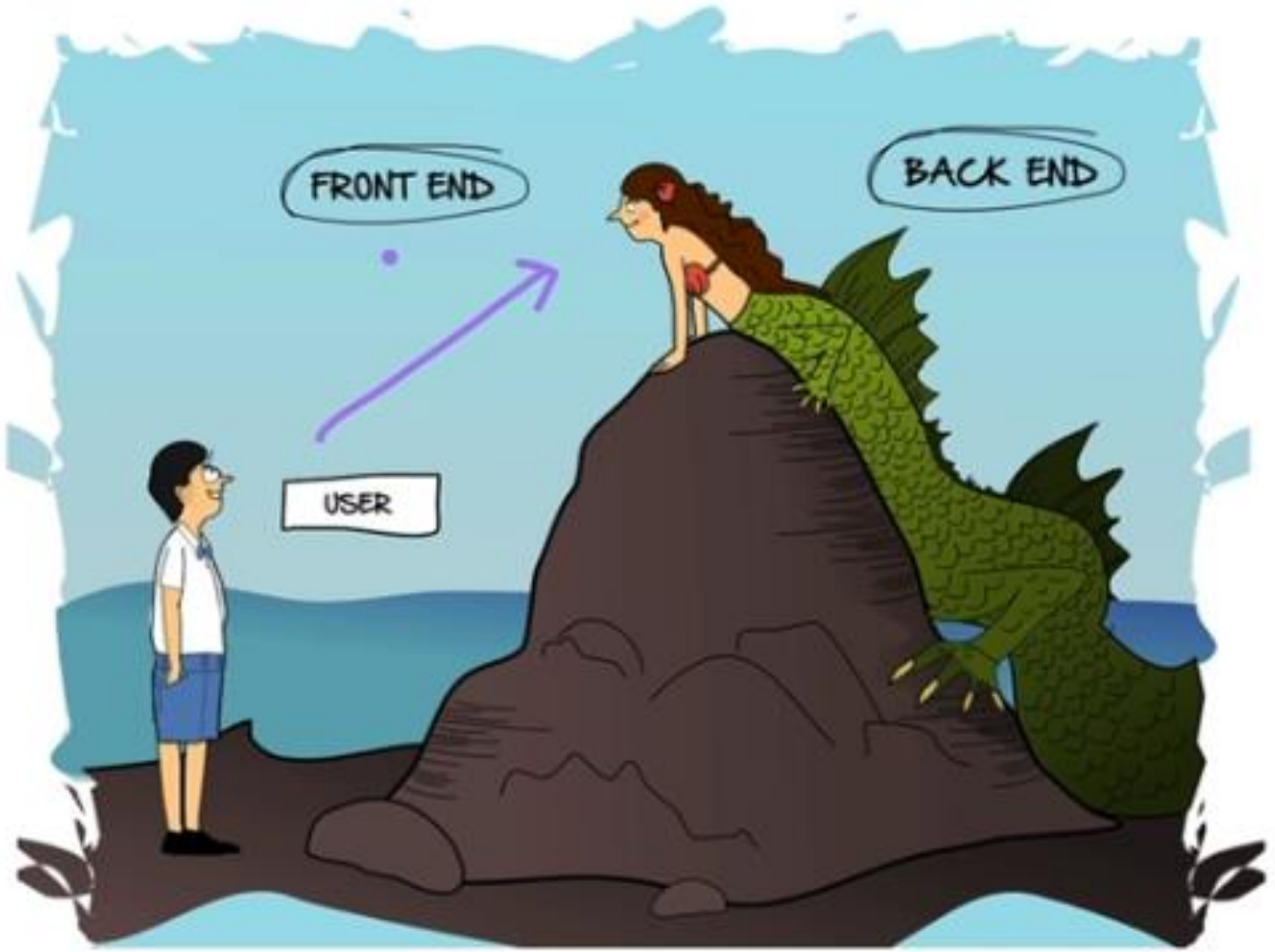
The Backend

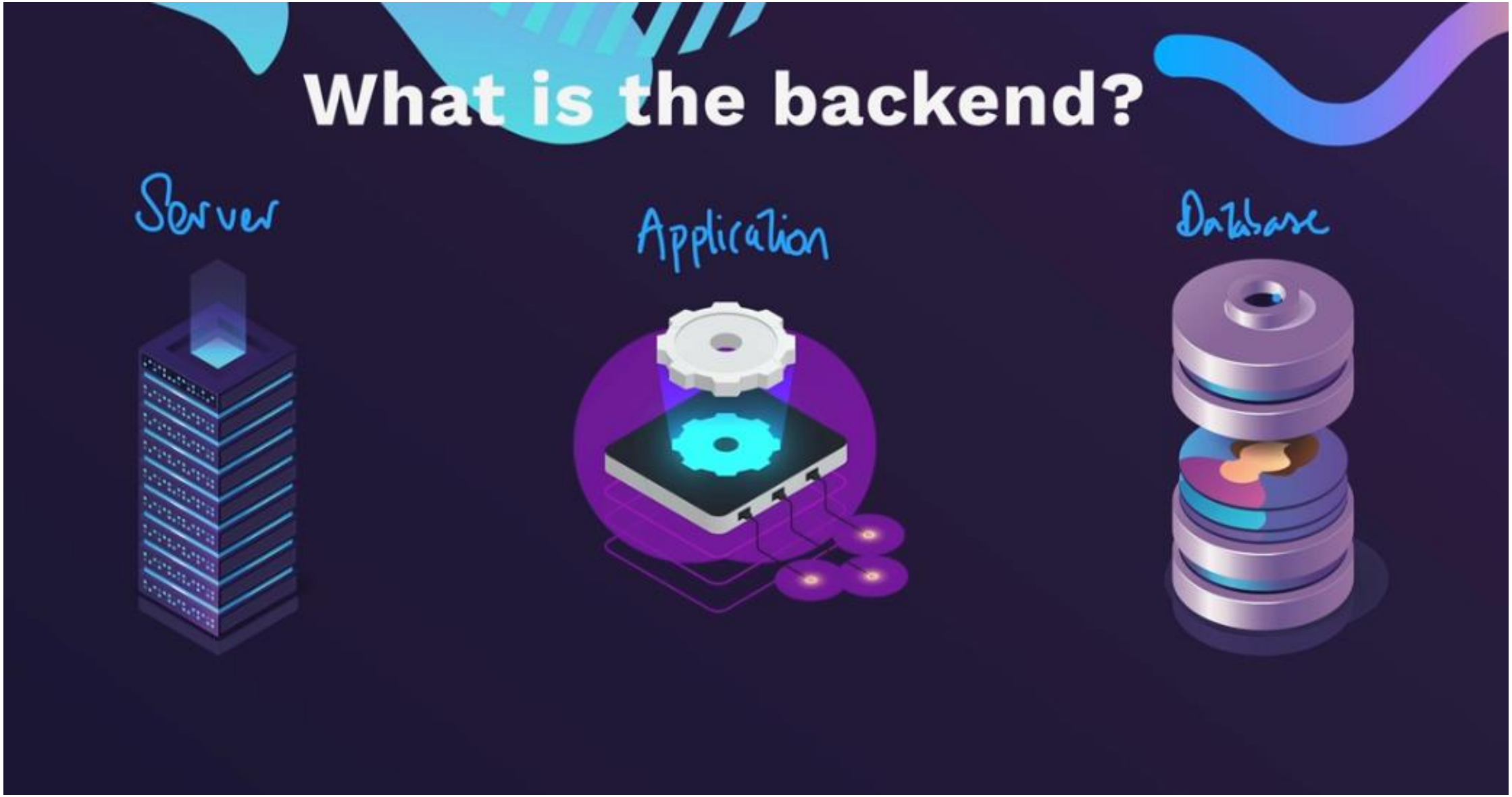
Behind the scenes code: Servers, Applications, and the Database



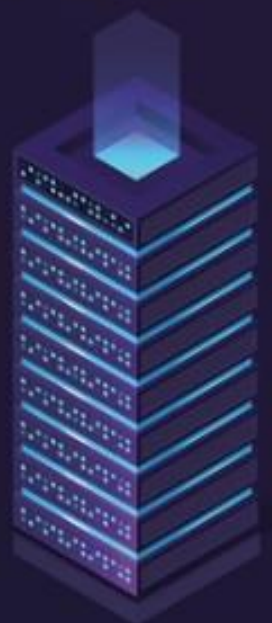
Full-Stack = Front-end + Back-end


FRONT END
BACK END
USER

What is the backend?



Server



Computer

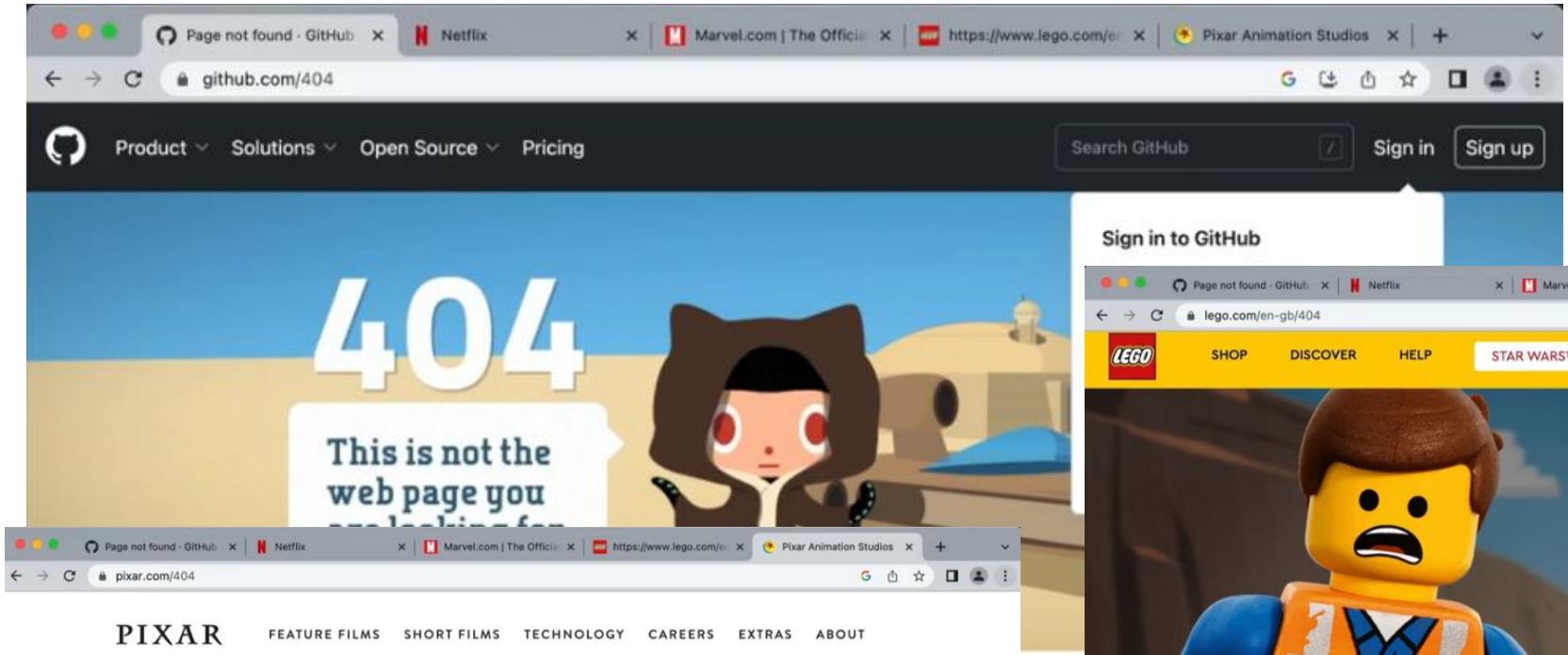
On 24/7

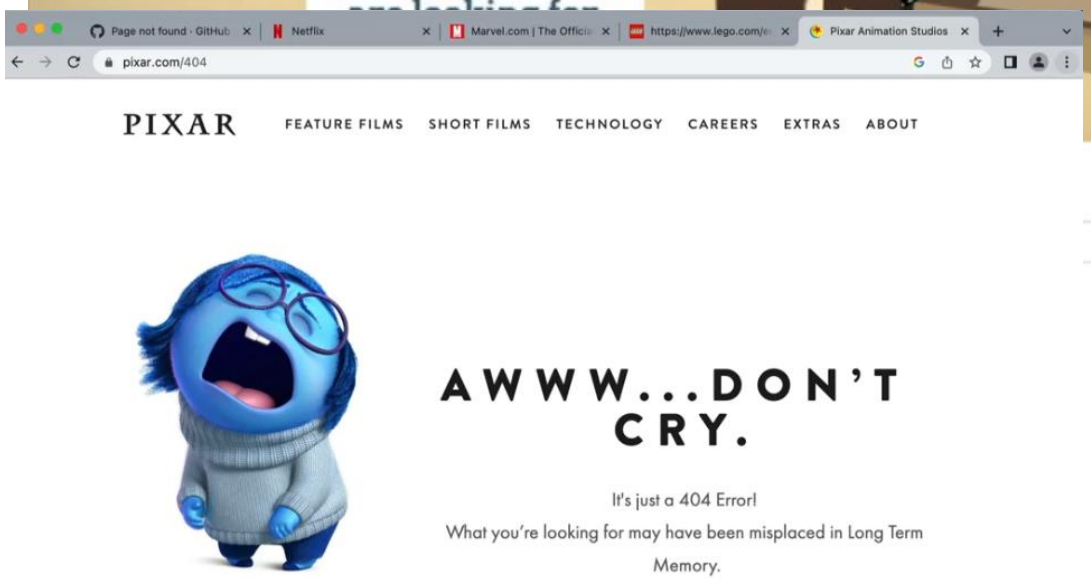
Application

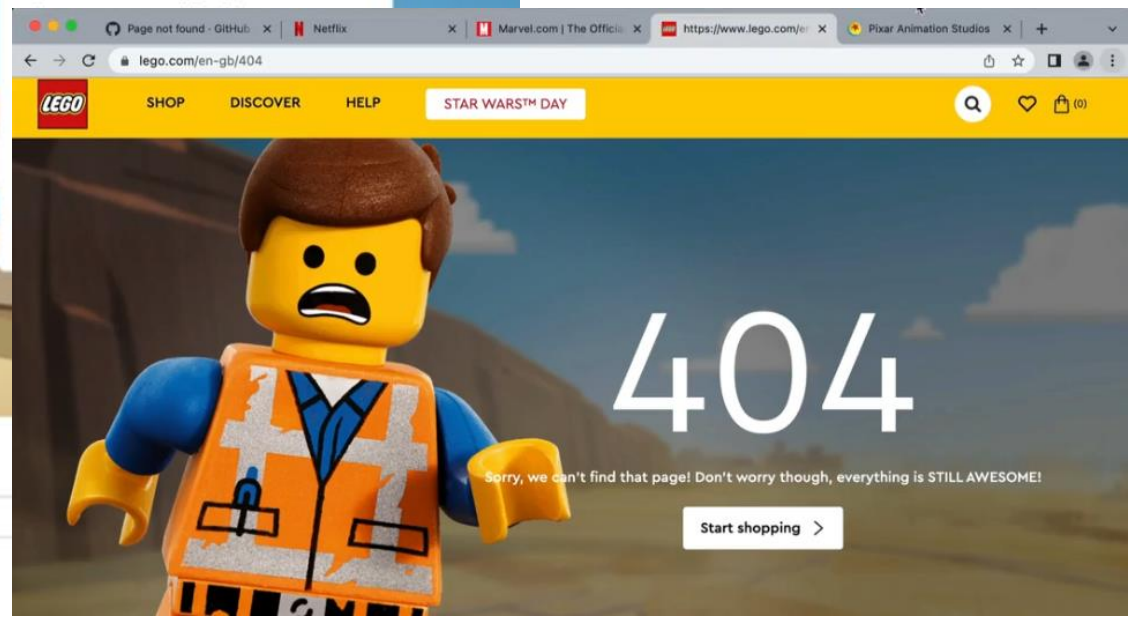


Logic

Respond to Requests


Page not found · GitHub
Netflix
Marvel.com | The Officia
https://www.lego.com/e
Pixar Animation Studios
github.com/404
Product
Solutions
Open Source
Pricing
Search GitHub
Sign in
Sign up
Sign in to GitHub
404
This is not the
web page you


Page not found · GitHub
Netflix
Marvel.com | The Officia
https://www.lego.com/e
Pixar Animation Studios
pixar.com/404
PIXAR
FEATURE FILMS
SHORT FILMS
TECHNOLOGY
CAREERS
EXTRAS
ABOUT
AWWW...DON'T CRY.
It's just a 404 Error!
What you're looking for may have been misplaced in Long Term Memory.

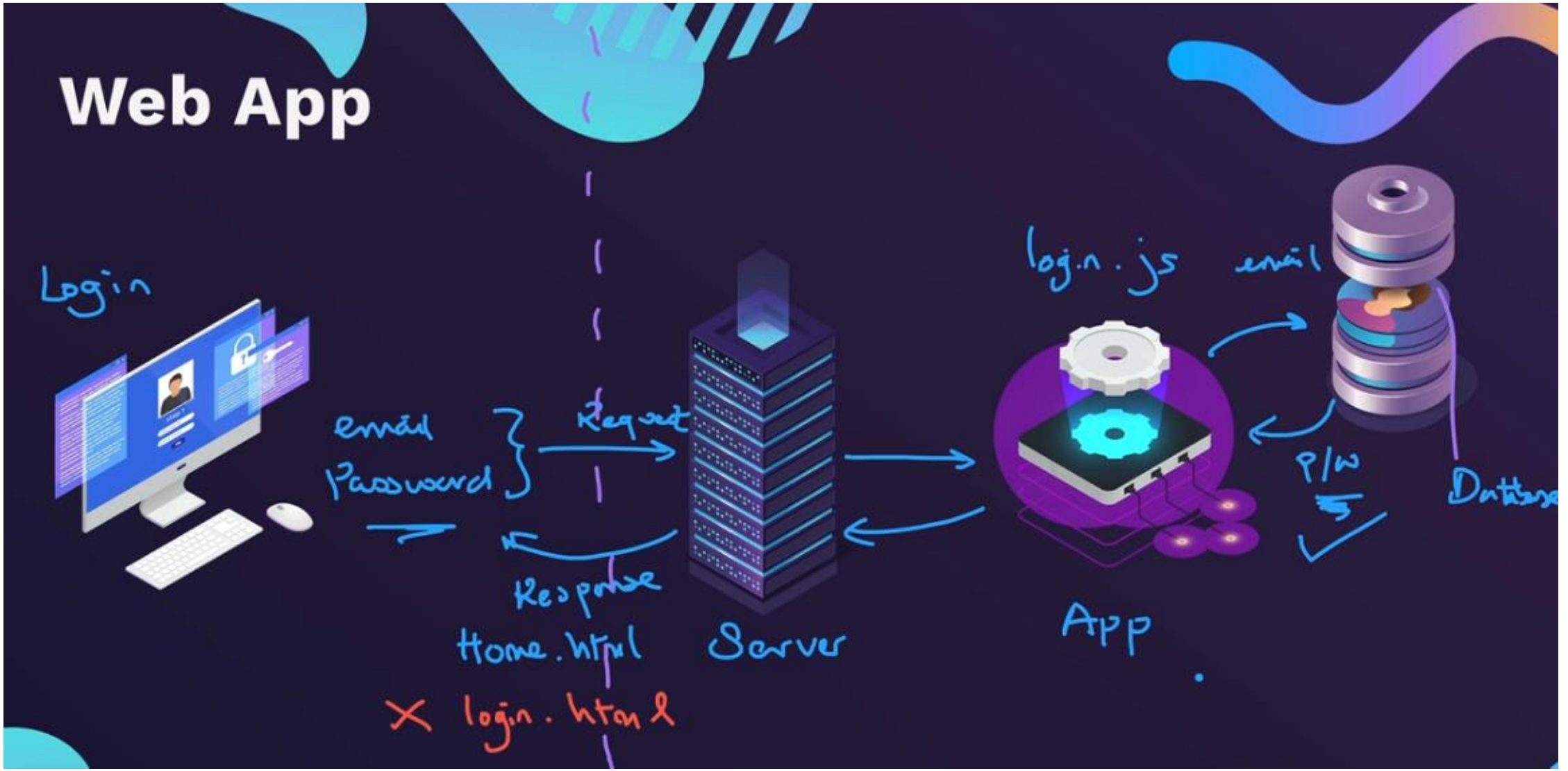

Page not found · GitHub
Netflix
Marvel.com | The Officia
https://www.lego.com/e
Pixar Animation Studios
lego.com/en-gb/404
LEGO
SHOP
DISCOVER
HELP
STAR WARS™ DAY
404
Sorry, we can't find that page! Don't worry though, everything is STILL AWESOME!
Start shopping

Database



Store Data


Static
Web Page
Front end
HTML
CSS
JS
Loading...
URL
Server


Web App
Login
email
Password
Request
Response
Home.html
login.html
Server
login.js
email
P/W
App

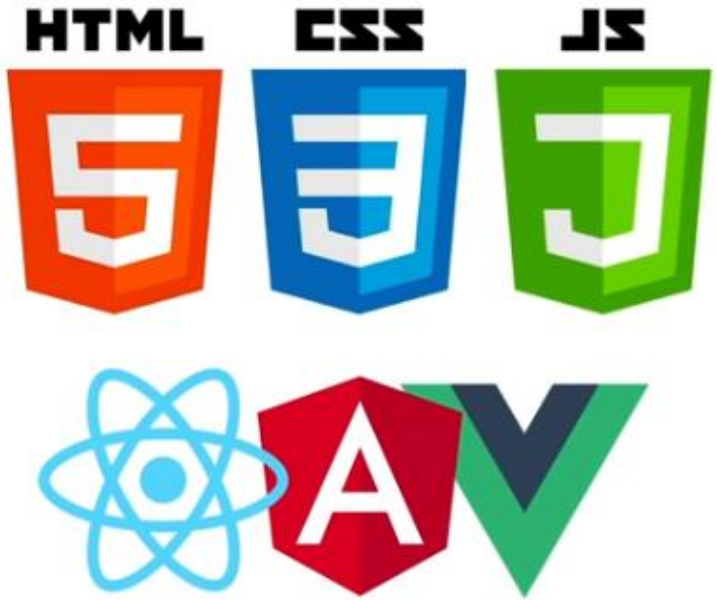

Restaurant
Kitchen
Pantry

Back-end Technologies

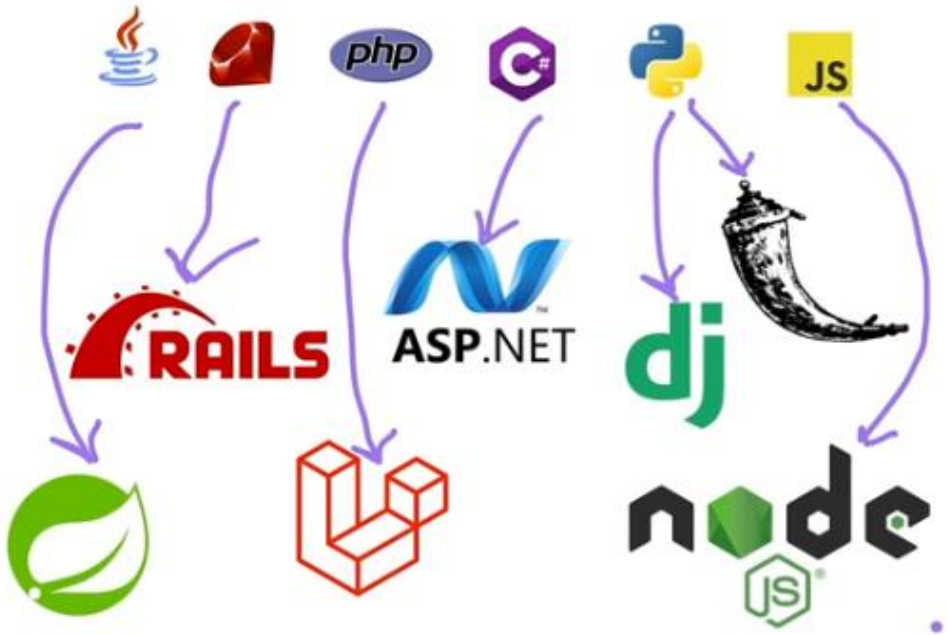
The tools used to build modern web applications.



Front-end



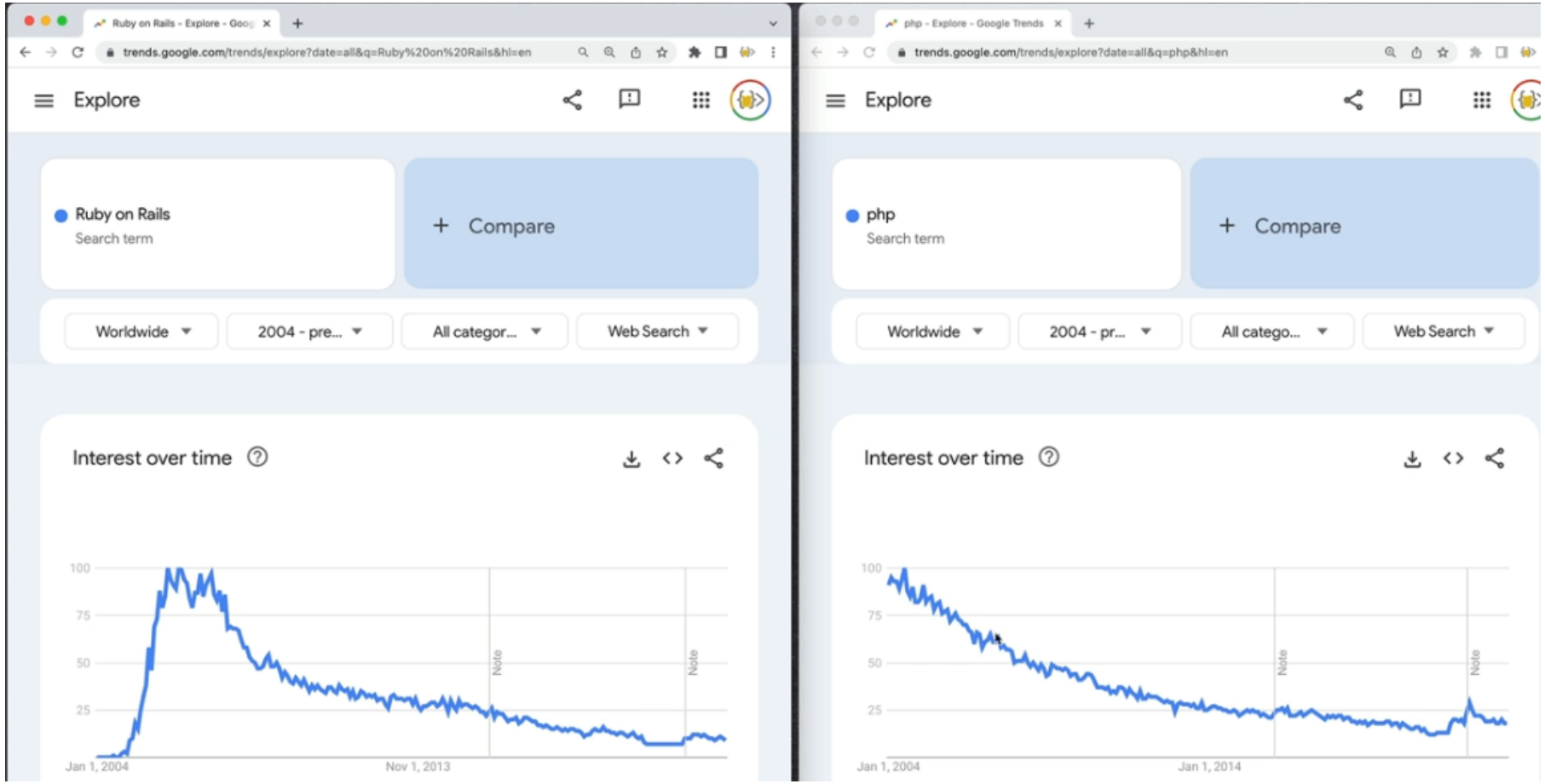
Back-end




node
JS
®


Professional Developers
Learning to Code
Node.js 46.31%
React.js 44.31%
jQuery 29.21%
Express 23.19%
Angular 23.06%
ASP.NET Core 20.71%
Vue.js 19.9%
ASP.NET 16.15%
Next.js 13.93%
Django 13.59%
Flask 13.24%
Laravel 10.3%
Angular.js 9.94%
Ruby on Rails 6.37%
FastAPI 6.01%
Blazor 4.92%
Svelte 4.11%


All Respondents
Professional Developers
Learning to Code
53,421 responses
JavaScript 67.9%
HTML/CSS 54.93%
SQL 52.64%
Python 43.51%


Ruby on Rails - Explore - Goog
trends.google.com/trends/explore?date=all&q=Ruby%20on%20Rails&hl=en
Explore
Ruby on Rails
Search term
Compare
Worldwide
2004 - pre...
All categor...
Web Search
Interest over time
100
75
50
25
Note
Note
Jan 1, 2004
Nov 1, 2013
php - Explore - Google Trends
trends.google.com/trends/explore?date=all&q=php&hl=en
Explore
php
Search term
Compare
Worldwide
2004 - pr...
All catego...
Web Search
Interest over time
100
75
50
25
Note
Note
Jan 1, 2004
Jan 1, 2014

What is Node.js?

Understand the runtime and the advantages of Node.js



Why use Frameworks?



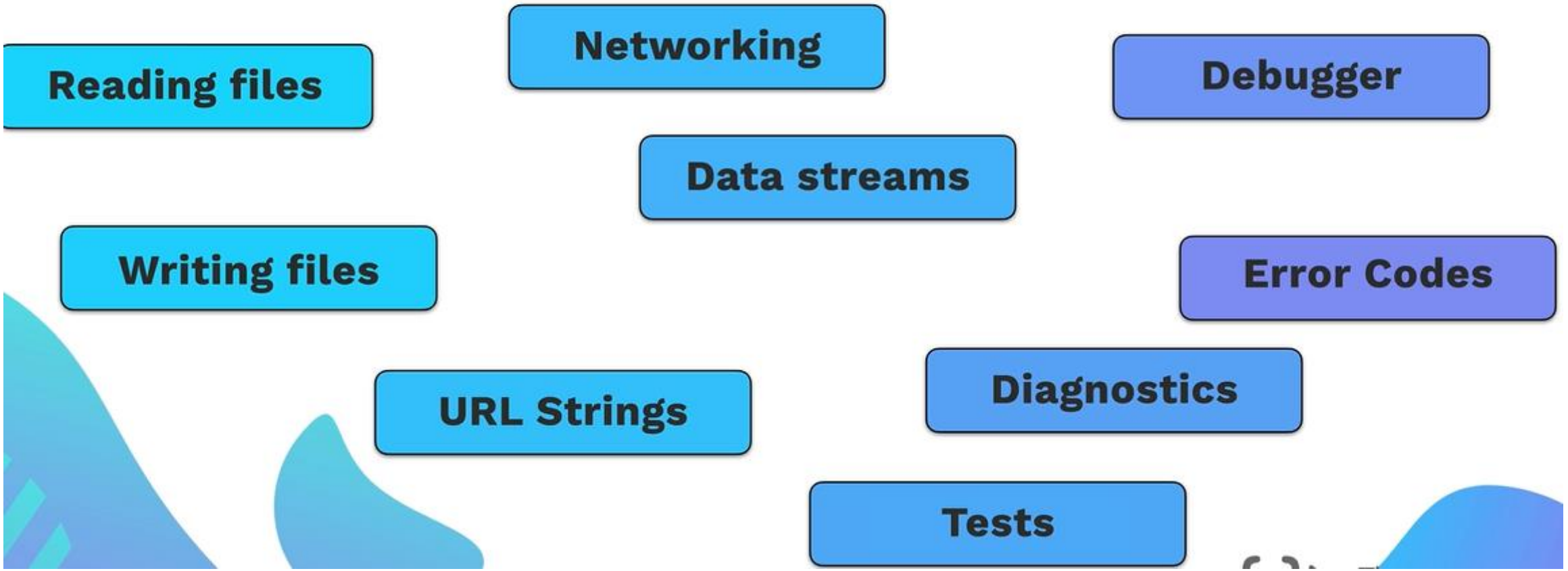


Why use Frameworks?





Reuse components. Reduce overhead



“An asynchronous event-driven JavaScript runtime, Node.js is designed to build scalable network applications.”

- According to Node.js



“An asynchronous event-driven JavaScript runtime, Node.js is designed to build scalable network applications.”



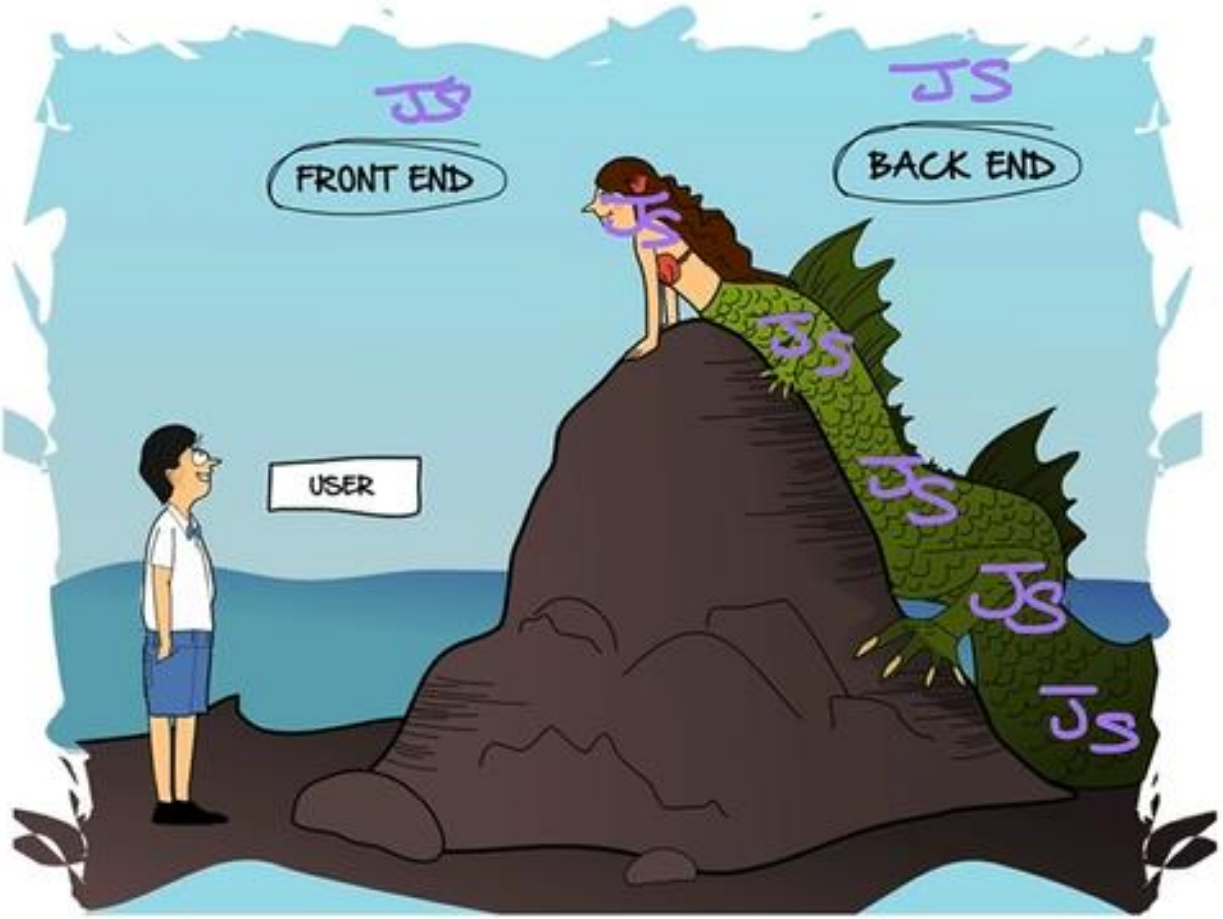

JS





JS

Why Node?



benefit



JS Fullstack



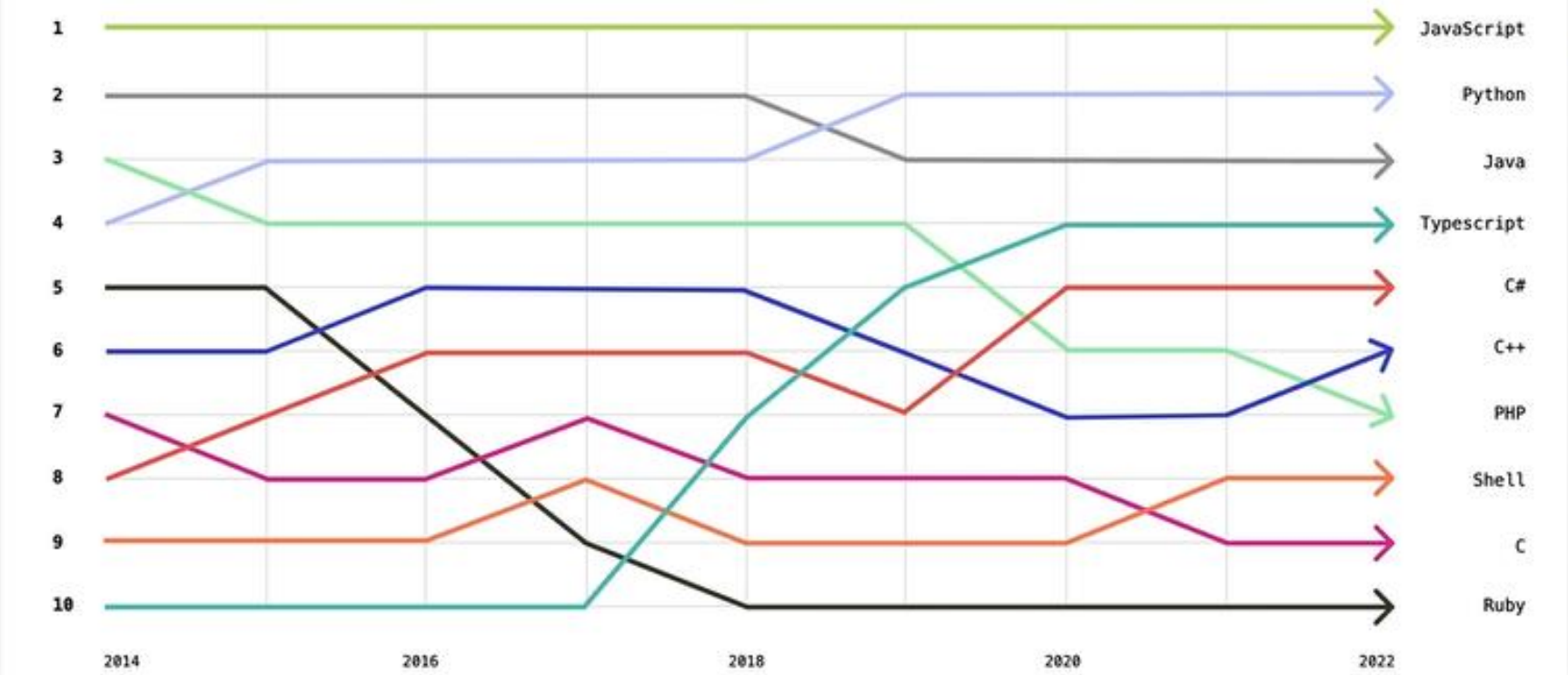
Scales



Non-blocking



Ecosystem


1
2
3
4
5
6
7
8
9
10
JavaScript
Python
Java
Typescript
C#
C++
PHP
Shell
C
Ruby
2014
2016
2018
2020
2022

Install Node.js on Windows

Installation Steps

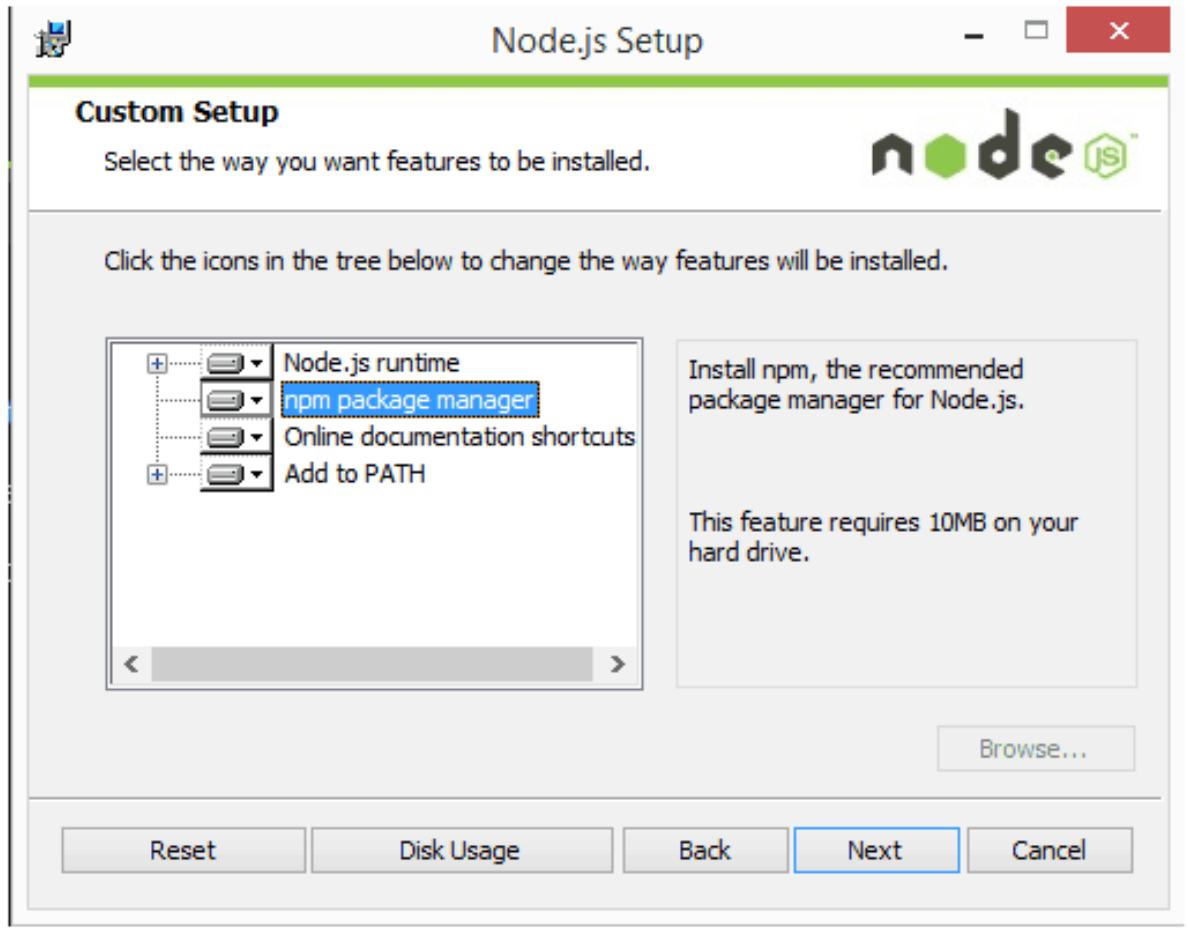
1. **Download the Windows installer from the Nodes.js® web site.**
2. **Choose the LTS version** that's shown on the left.



3. **Run the installer** (the .msi file you downloaded in the previous step.)

3. **Run the installer** (the .msi file you downloaded in the previous step.)

4. **Follow the prompts in the installer** (Accept the license agreement, click the NEXT button a bunch of times and accept the default installation settings).



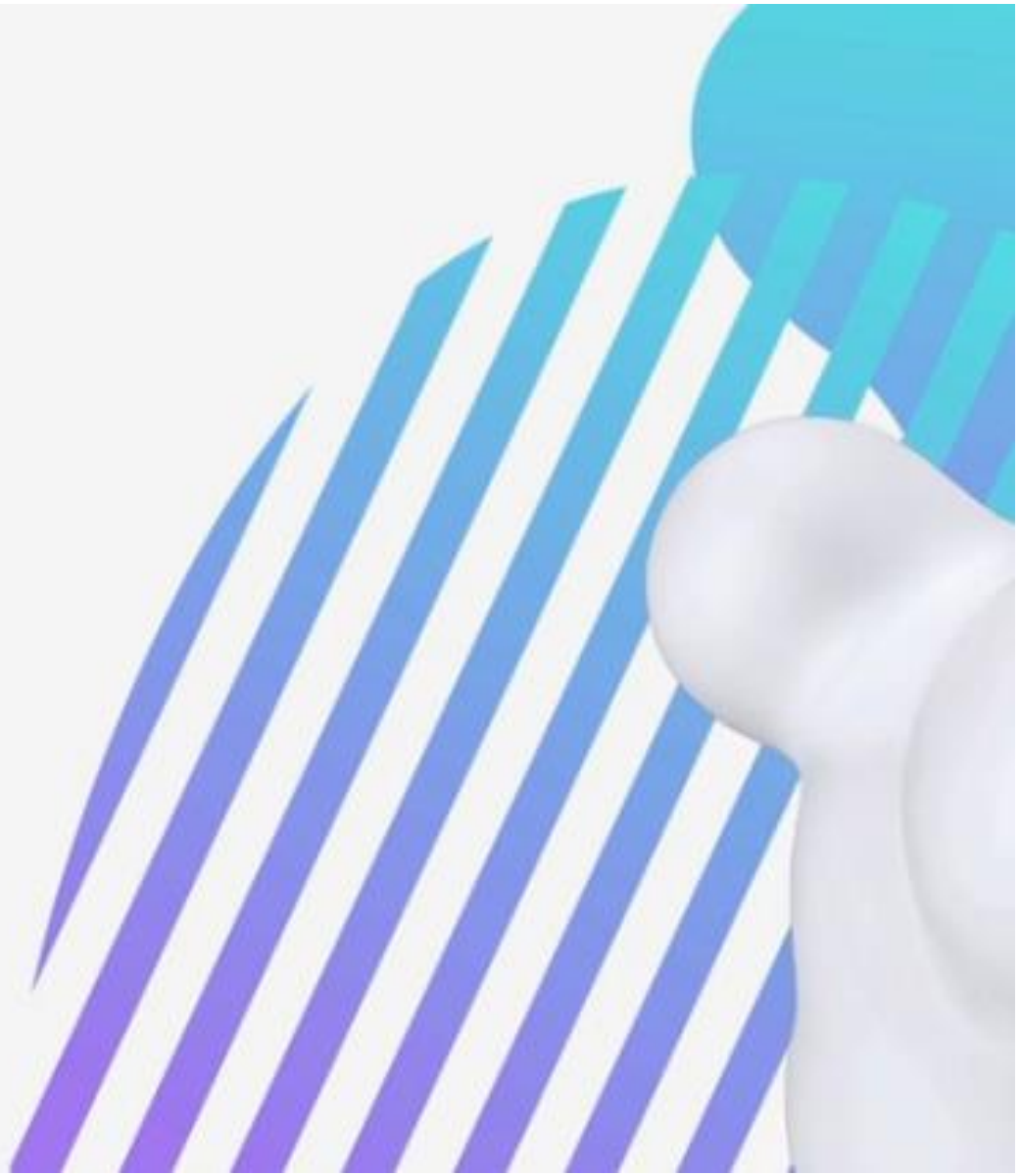
5. **Restart your computer.** You won't be able to run Node.js® until you restart your computer.

6. Confirm that Node has been installed successfully on your computer by opening a Hyper terminal and typing in the commands `node --version`

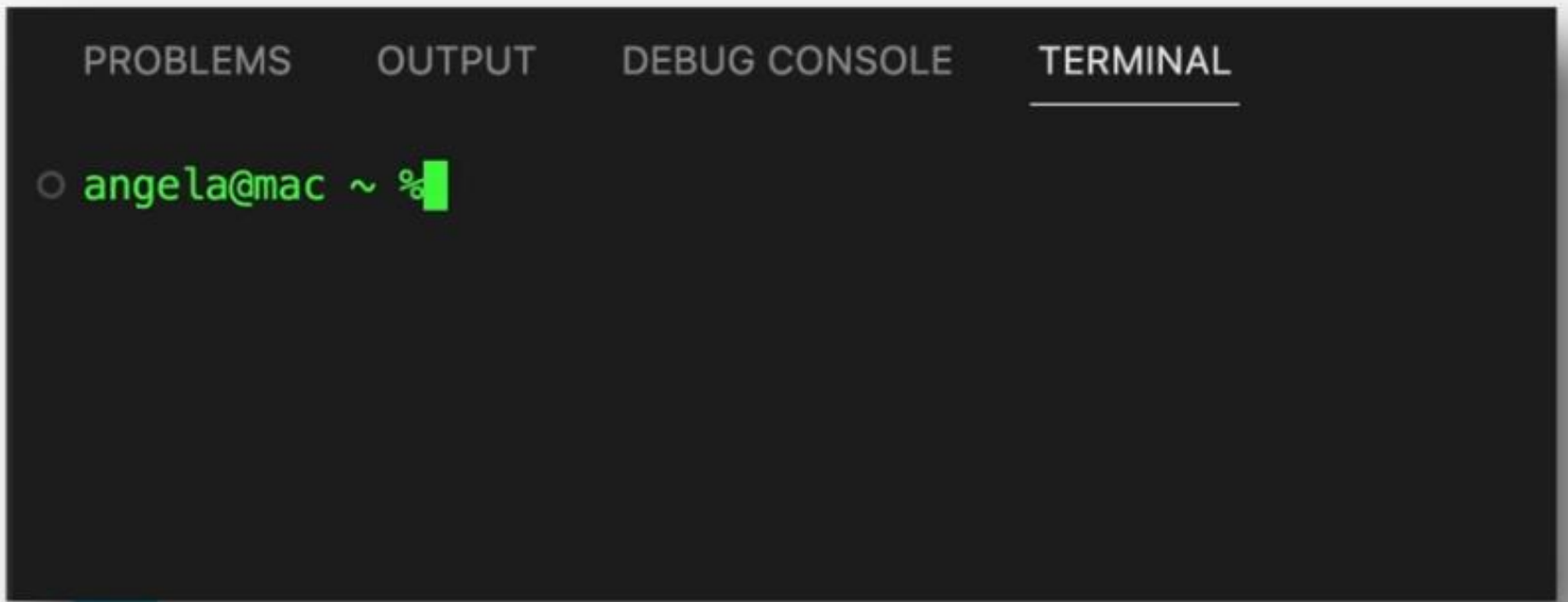
You should see the version of node you just installed.

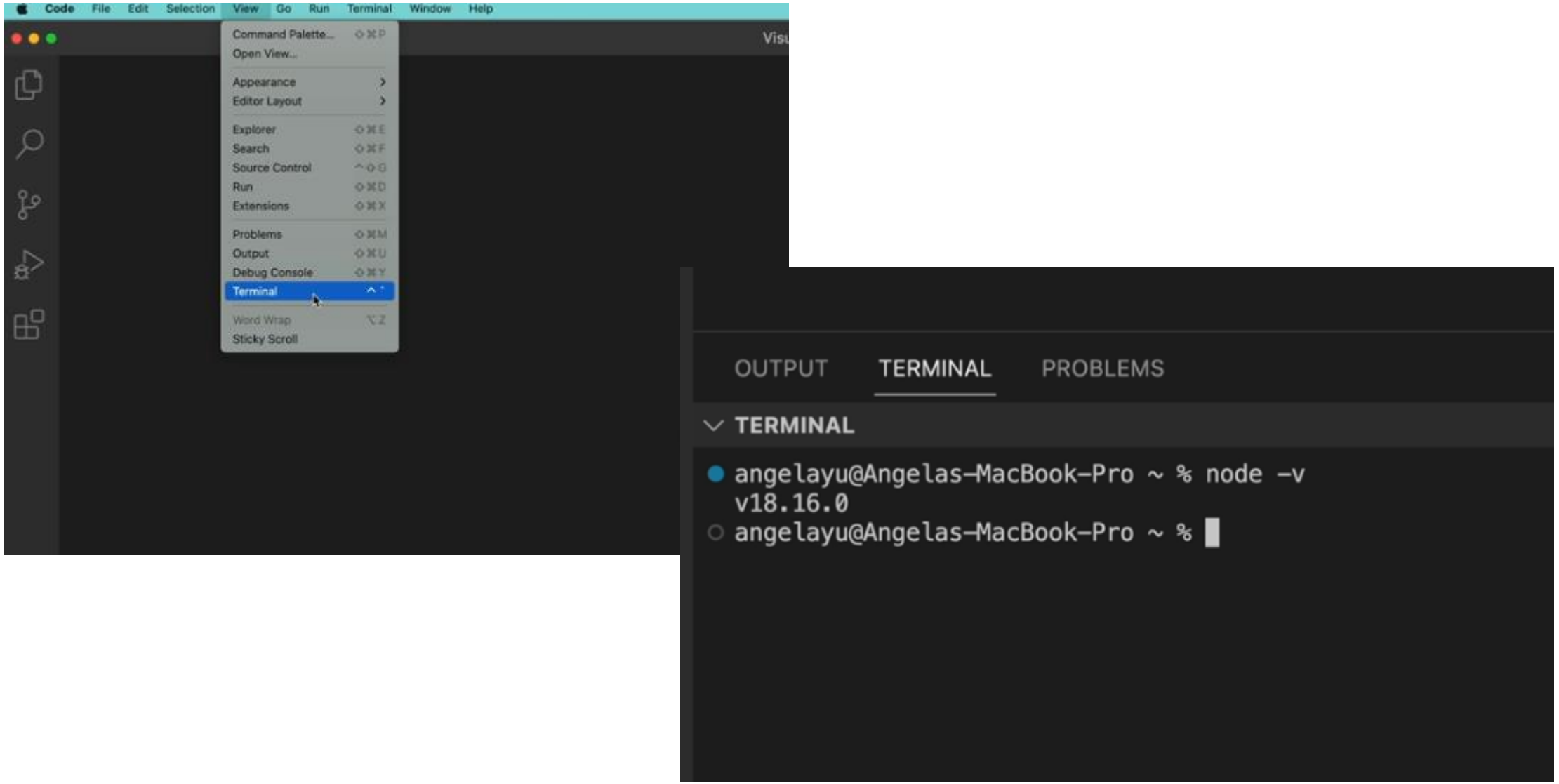
Using Node

Versioning, REPL and running code using Node.



Version Check




Code File Edit Selection View Go Run Terminal Window Help
Command Palette...
Open View...
Appearance
Editor Layout
Explorer
Search
Source Control
Run
Extensions
Problems
Output
Debug Console
Terminal
Word Wrap
Sticky Scroll
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
angelayu@Angelas-MacBook-Pro ~ % node -v
v18.16.0
angelayu@Angelas-MacBook-Pro ~ %

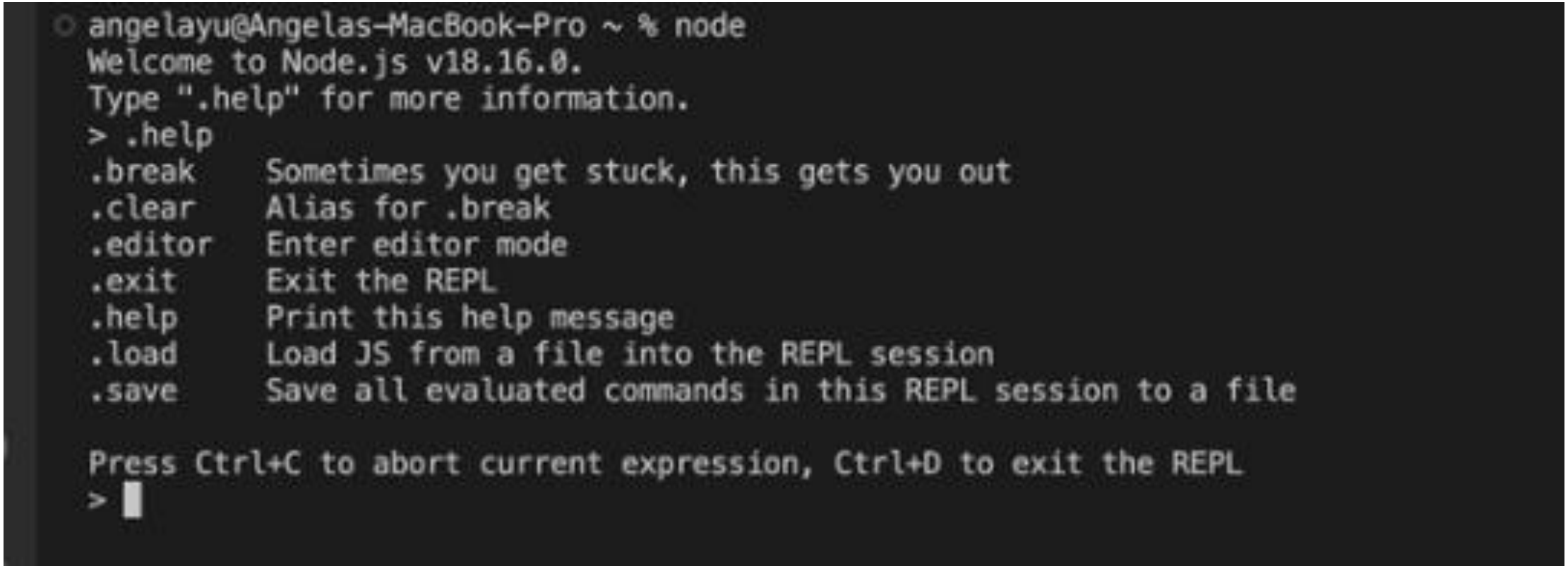
Node Read Eval Print Loop

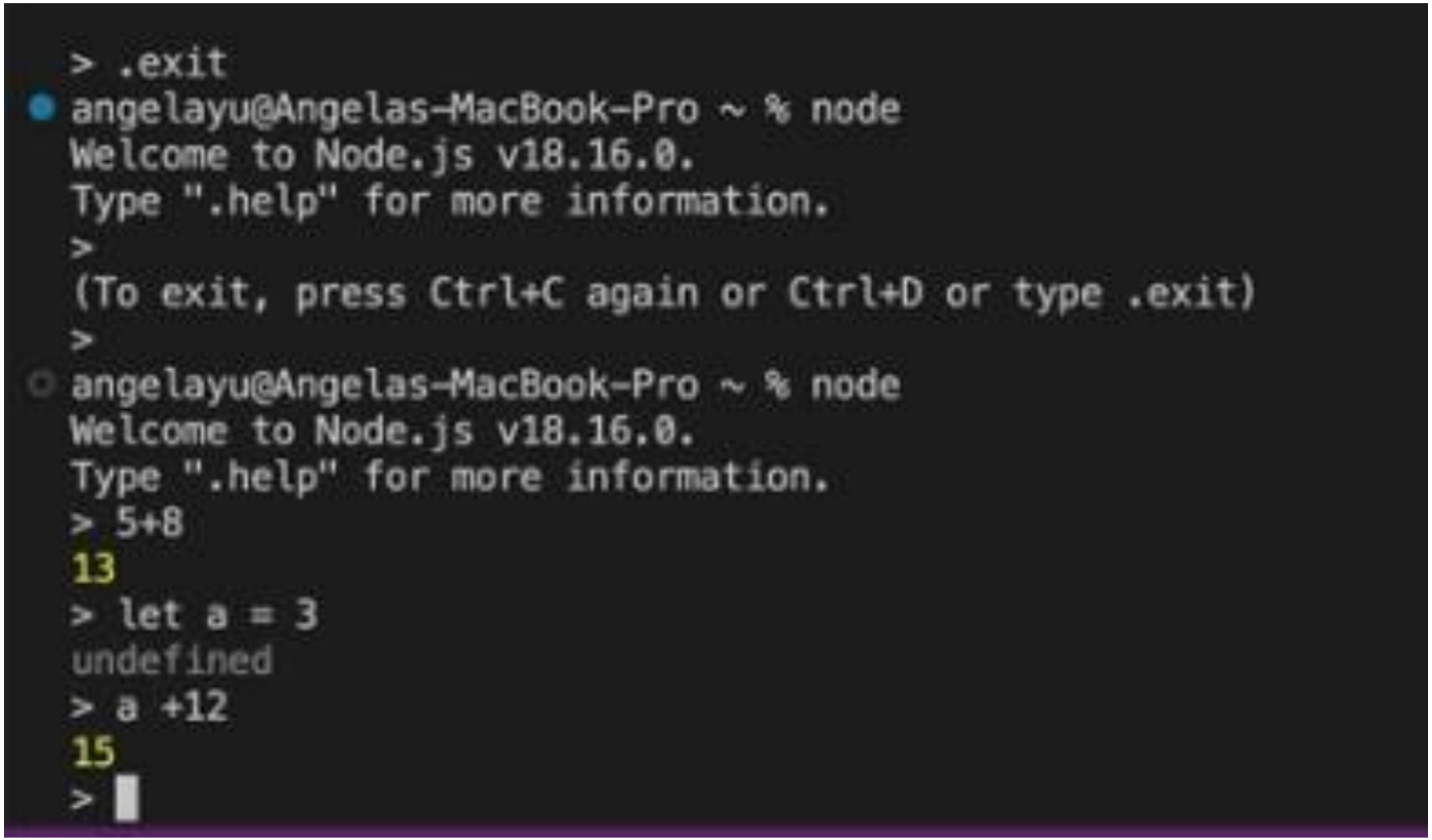


is a computer environment where user inputs are read and evaluated, and then the results are returned to the user.



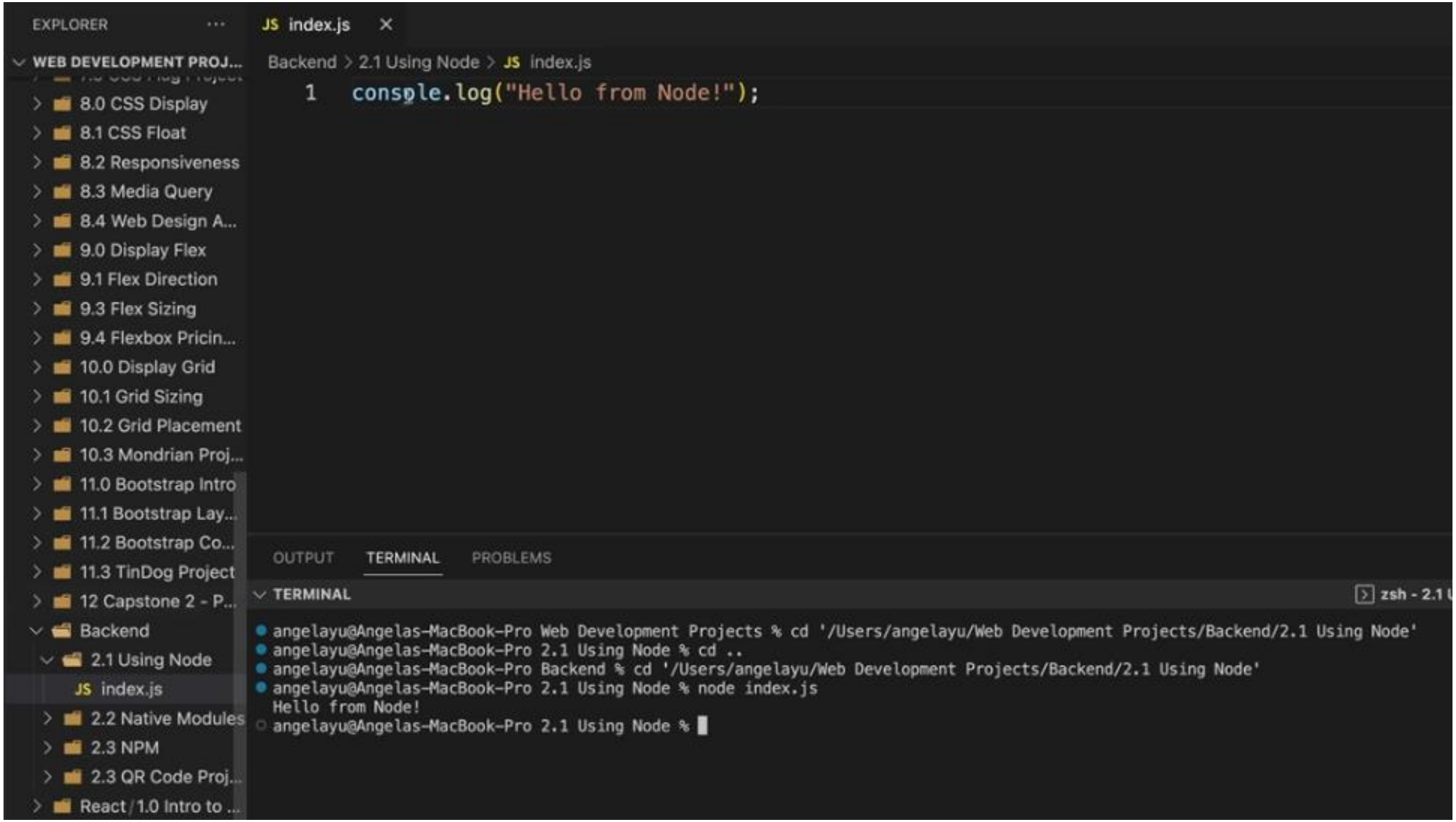
Node REPL


angelayu@Angelas-MacBook-Pro ~ % node
Welcome to Node.js v18.16.0.
Type ".help" for more information.
> .help
.break Sometimes you get stuck, this gets you out
.clear Alias for .break
.editor Enter editor mode
.exit Exit the REPL
.help Print this help message
.load Load JS from a file into the REPL session
.save Save all evaluated commands in this REPL session to a file
Press Ctrl+C to abort current expression, Ctrl+D to exit the REPL
>


> .exit
angelayu@Angelas-MacBook-Pro ~ % node
Welcome to Node.js v18.16.0.
Type ".help" for more information.
>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
>
angelayu@Angelas-MacBook-Pro ~ % node
Welcome to Node.js v18.16.0.
Type ".help" for more information.
> 5+8
13
> let a = 3
undefined
> a +12
15
>

Running JS Files

```
node index.js
```


EXPLORER
index.js
WEB DEVELOPMENT PROJ...
8.0 CSS Display
8.1 CSS Float
8.2 Responsiveness
8.3 Media Query
8.4 Web Design A...
9.0 Display Flex
9.1 Flex Direction
9.3 Flex Sizing
9.4 Flexbox Pricin...
10.0 Display Grid
10.1 Grid Sizing
10.2 Grid Placement
10.3 Mondrian Proj...
11.0 Bootstrap Intro
11.1 Bootstrap Lay...
11.2 Bootstrap Co...
11.3 TinDog Project
12 Capstone 2 - P...
Backend
2.1 Using Node
index.js
2.2 Native Modules
2.3 NPM
2.3 QR Code Proj...
React / 1.0 Intro to ...
Backend > 2.1 Using Node > index.js
1 console.log("Hello from Node!");
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
zsh - 2.1
angelayu@Angelas-MacBook-Pro Web Development Projects % cd '/Users/angelayu/Web Development Projects/Backend/2.1 Using Node'
angelayu@Angelas-MacBook-Pro 2.1 Using Node % cd ..
angelayu@Angelas-MacBook-Pro Backend % cd '/Users/angelayu/Web Development Projects/Backend/2.1 Using Node'
angelayu@Angelas-MacBook-Pro 2.1 Using Node % node index.js
Hello from Node!
angelayu@Angelas-MacBook-Pro 2.1 Using Node %

Native Node Modules

Leveraging Node

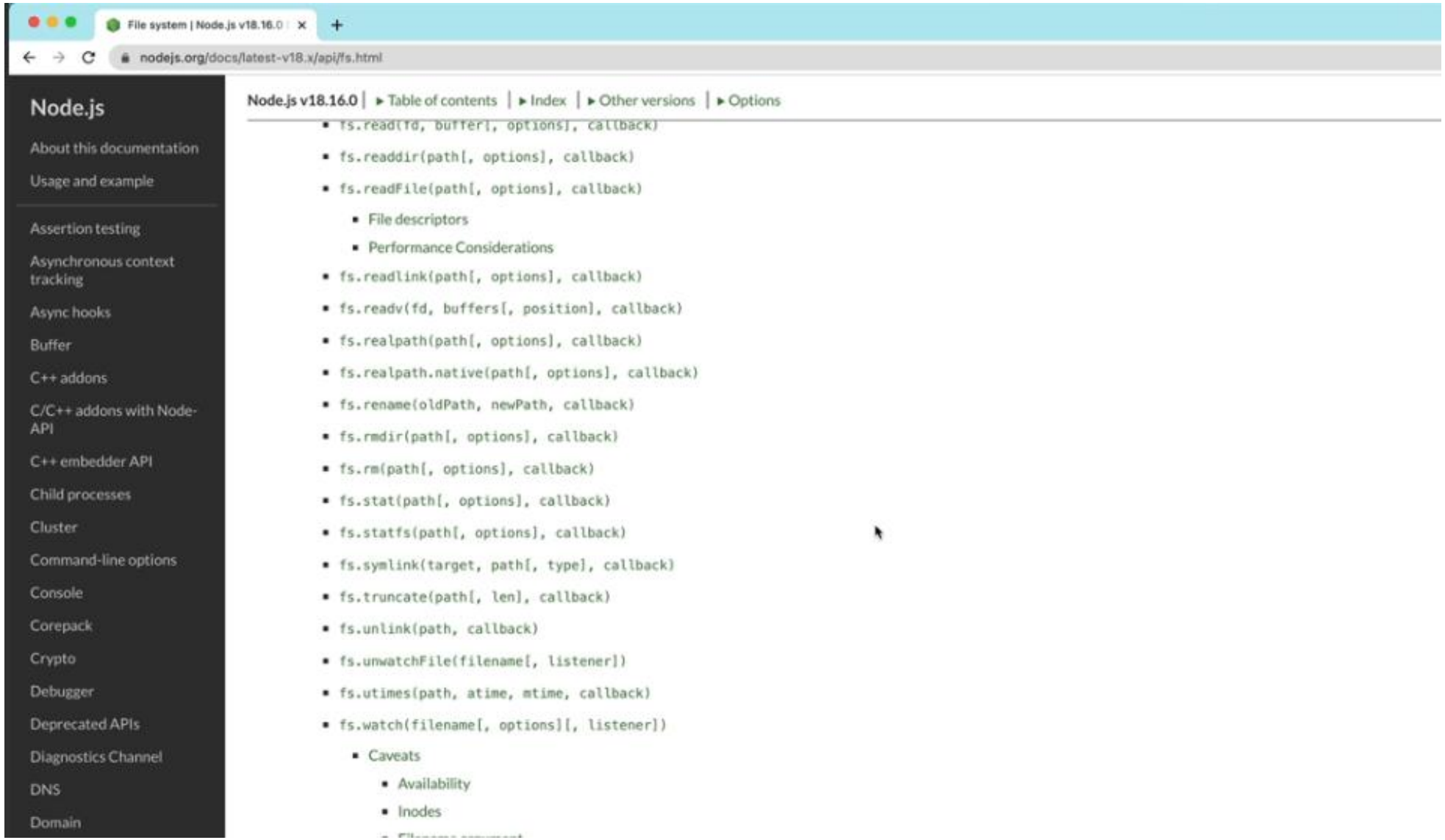


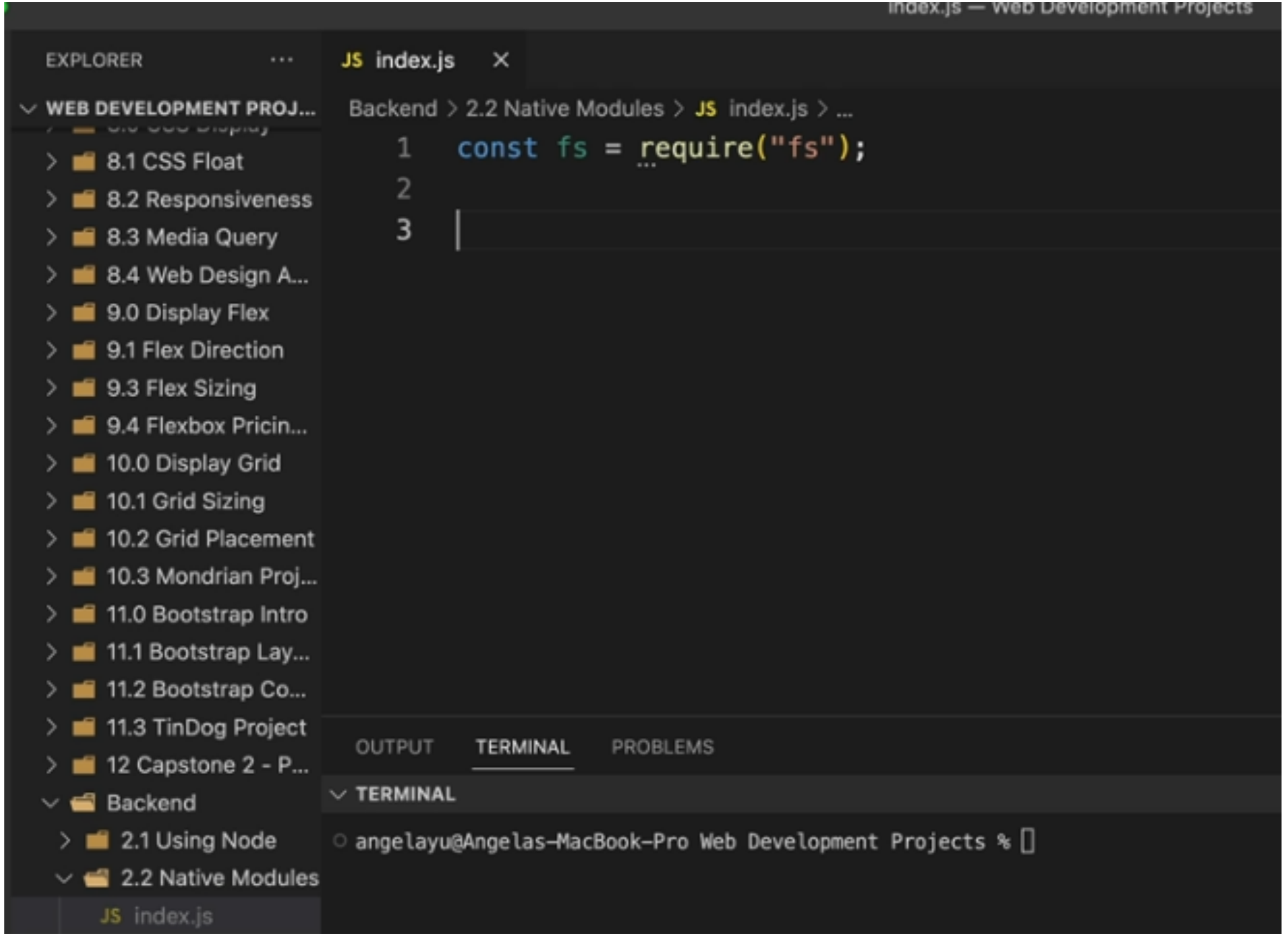
What are Native Modules?

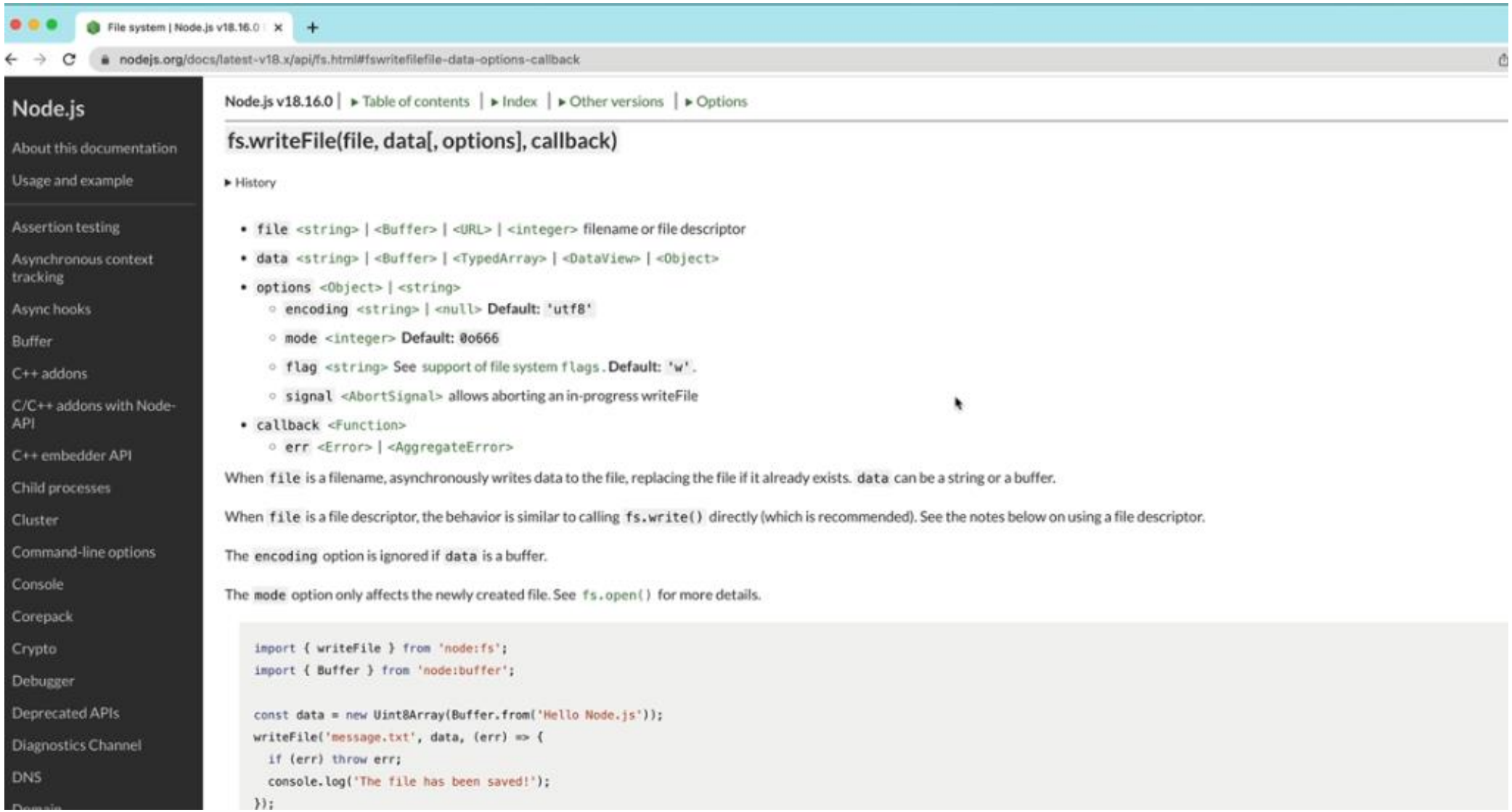


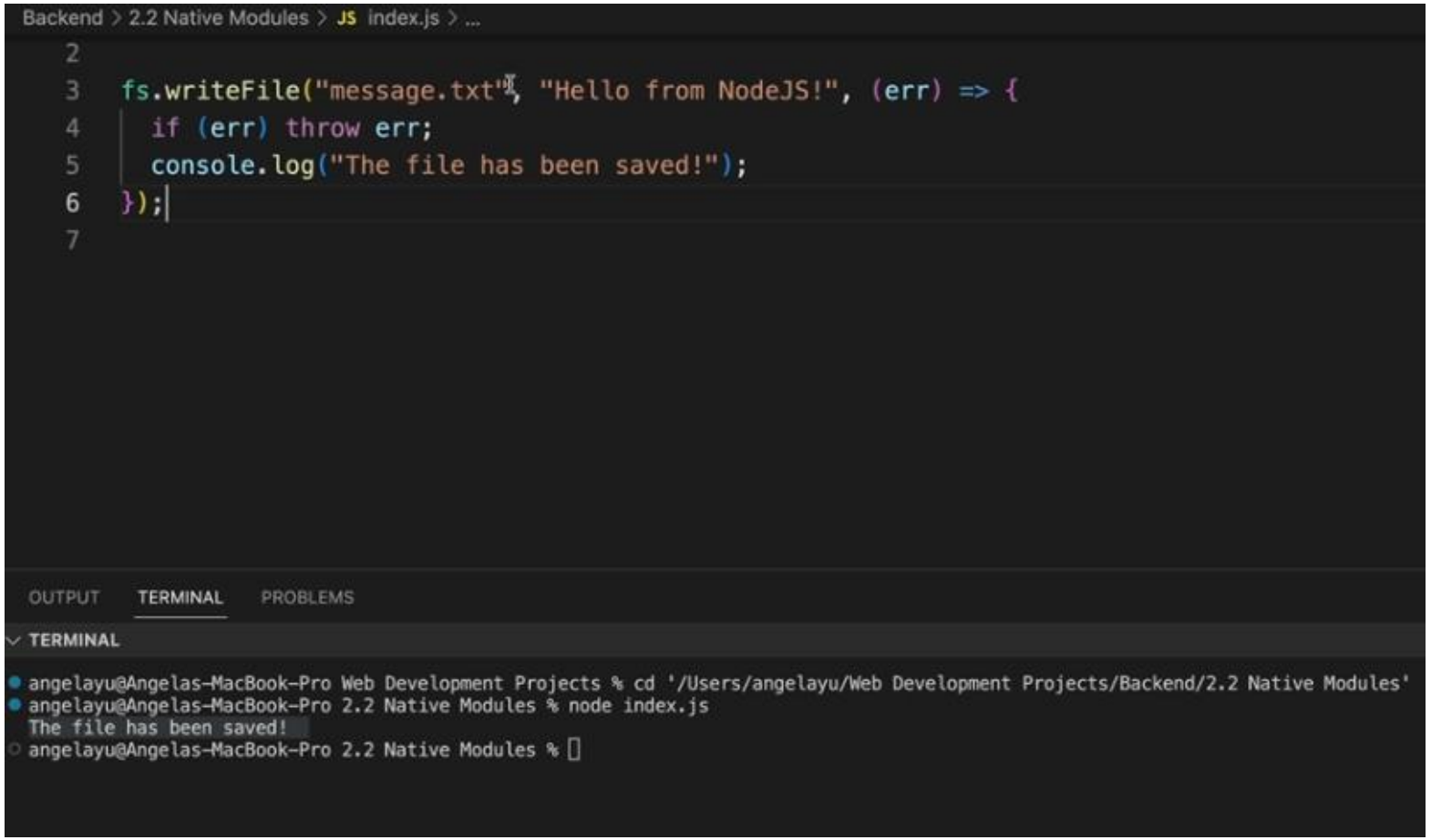
Node

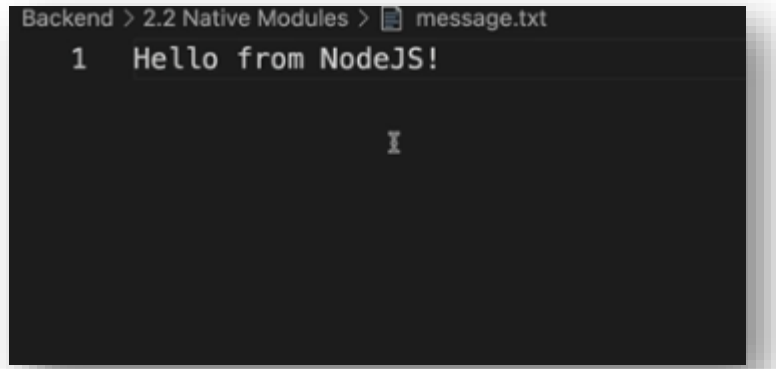
- App
  - ↳ File
  - ↳ Network

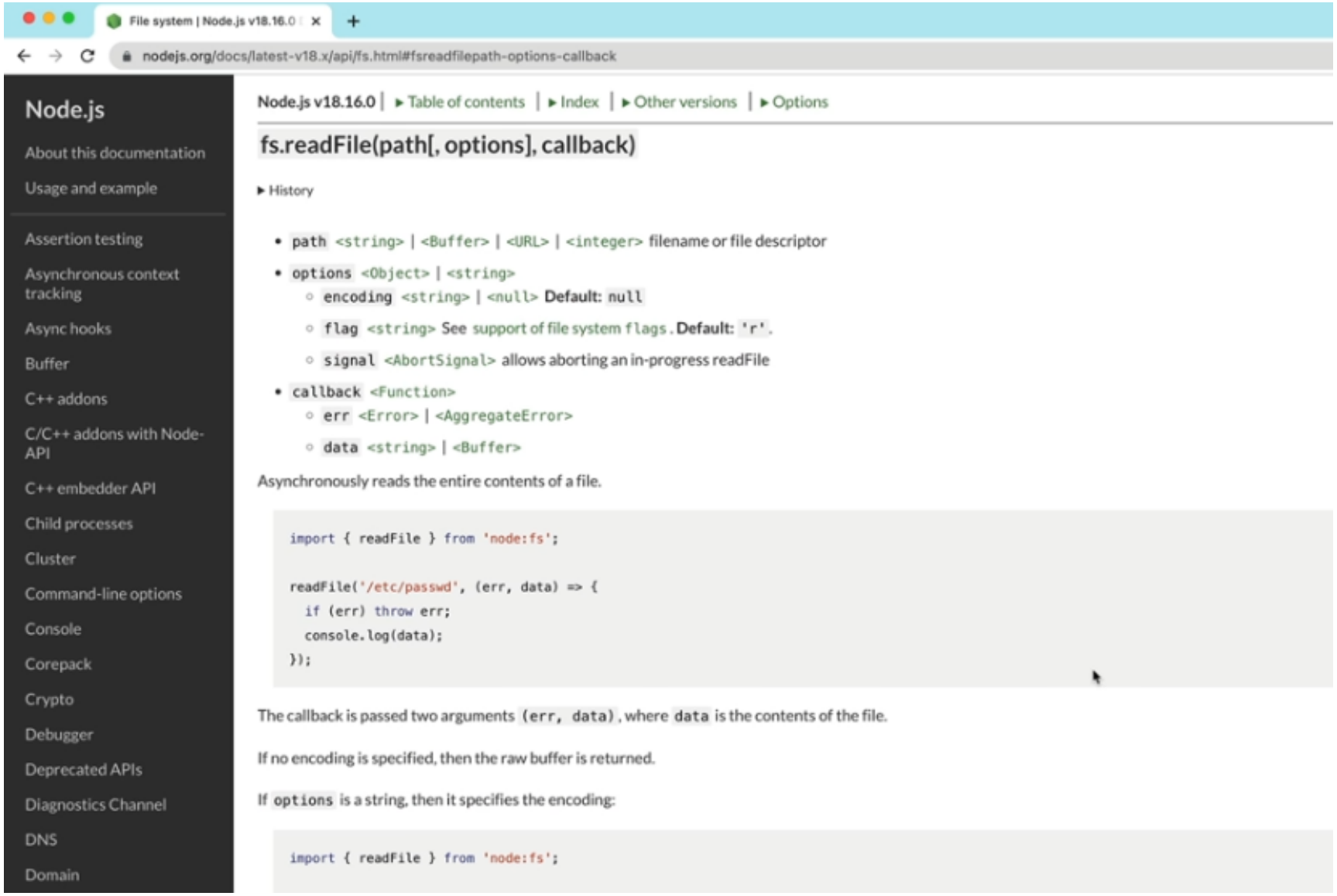

File system | Node.js v18.16.0
nodejs.org/docs/latest-v18.x/api/fs.html
Node.js
About this documentation
Usage and example
Assertion testing
Asynchronous context tracking
Async hooks
Buffer
C++ addons
C/C++ addons with Node-API
C++ embedder API
Child processes
Cluster
Command-line options
Console
Corepack
Crypto
Debugger
Deprecated APIs
Diagnostics Channel
DNS
Domain
Node.js v18.16.0 | ▸ Table of contents | ▸ Index | ▸ Other versions | ▸ Options
fs.read(fd, buffer[, options], callback)
fs.readdir(path[, options], callback)
fs.readFile(path[, options], callback)
File descriptors
Performance Considerations
fs.readlink(path[, options], callback)
fs.readv(fd, buffers[, position], callback)
fs.realpath(path[, options], callback)
fs.realpath.native(path[, options], callback)
fs.rename(oldPath, newPath, callback)
fs.rmdir(path[, options], callback)
fs.rm(path[, options], callback)
fs.stat(path[, options], callback)
fs.statfs(path[, options], callback)
fs.symlink(target, path[, type], callback)
fs.truncate(path[, len], callback)
fs.unlink(path, callback)
fs.unwatchFile(filename[, listener])
fs.utimes(path, atime, mtime, callback)
fs.watch(filename[, options][, listener])
Caveats
Availability
Inodes


index.js — Web Development Projects
EXPLORER
JS index.js
WEB DEVELOPMENT PROJ...
8.1 CSS Float
8.2 Responsiveness
8.3 Media Query
8.4 Web Design A...
9.0 Display Flex
9.1 Flex Direction
9.3 Flex Sizing
9.4 Flexbox Pricin...
10.0 Display Grid
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10.2 Grid Placement
10.3 Mondrian Proj...
11.0 Bootstrap Intro
11.1 Bootstrap Lay...
11.2 Bootstrap Co...
11.3 TinDog Project
12 Capstone 2 - P...
Backend
2.1 Using Node
2.2 Native Modules
JS index.js
Backend > 2.2 Native Modules > JS index.js > ...
1 const fs = require("fs");
2
3
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
angelayu@Angelas-MacBook-Pro Web Development Projects %

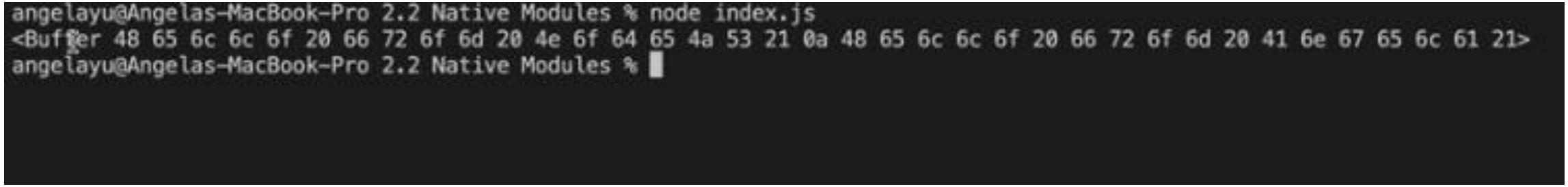

File system | Node.js v18.16.0
nodejs.org/docs/latest-v18.x/api/fs.html#fswritefilefile-data-options-callback
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Console
Corepack
Crypto
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Deprecated APIs
Diagnostics Channel
DNS
Node.js v18.16.0 | ▸ Table of contents | ▸ Index | ▸ Other versions | ▸ Options
fs.writeFile(file, data[, options], callback)
▸ History
file <string> | <Buffer> | <URL> | <integer> filename or file descriptor
data <string> | <Buffer> | <TypedArray> | <DataView> | <Object>
options <Object> | <string>
encoding <string> | <null> Default: 'utf8'
mode <integer> Default: 0o666
flag <string> See support of file system flags. Default: 'w'.
signal <AbortSignal> allows aborting an in-progress writeFile
callback <Function>
err <Error> | <AggregateError>
When file is a filename, asynchronously writes data to the file, replacing the file if it already exists. data can be a string or a buffer.
When file is a file descriptor, the behavior is similar to calling fs.write() directly (which is recommended). See the notes below on using a file descriptor.
The encoding option is ignored if data is a buffer.
The mode option only affects the newly created file. See fs.open() for more details.
import { writeFile } from 'node:fs';
import { Buffer } from 'node:buffer';
const data = new Uint8Array(Buffer.from('Hello Node.js'));
writeFile('message.txt', data, (err) => {
if (err) throw err;
console.log('The file has been saved!');
});

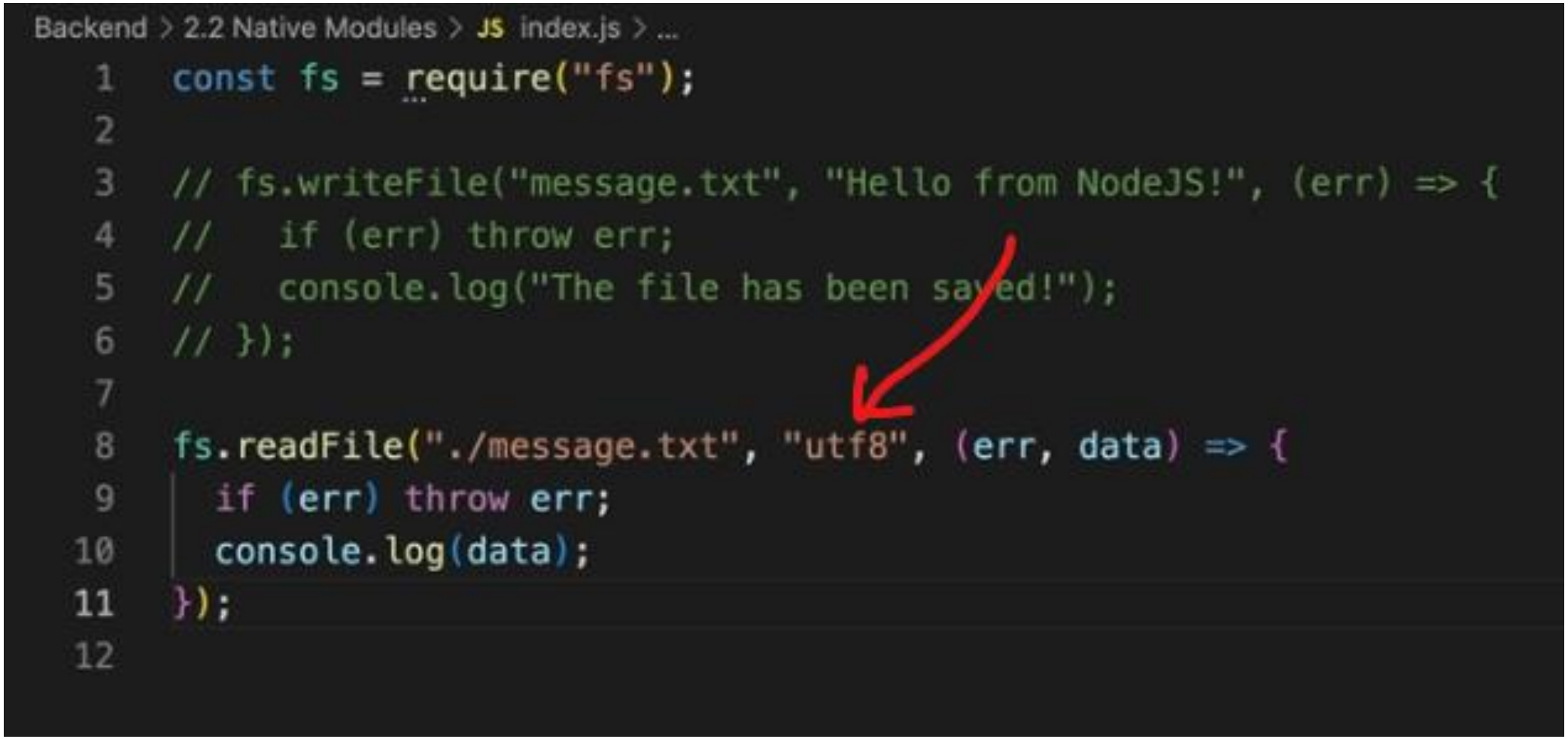

Backend > 2.2 Native Modules > JS index.js > ...
2
3 fs.writeFile("message.txt", "Hello from NodeJS!", (err) => {
4 if (err) throw err;
5 console.log("The file has been saved!");
6 });
7
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
angelayu@Angelas-MacBook-Pro Web Development Projects % cd '/Users/angelayu/Web Development Projects/Backend/2.2 Native Modules'
angelayu@Angelas-MacBook-Pro 2.2 Native Modules % node index.js
The file has been saved!
angelayu@Angelas-MacBook-Pro 2.2 Native Modules %


Backend > 2.2 Native Modules > message.txt
1 Hello from NodeJS!


File system | Node.js v18.16.0
nodejs.org/docs/latest-v18.x/api/fs.html#fsreadfilepath-options-callback
Node.js
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C++ addons
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C++ embedder API
Child processes
Cluster
Command-line options
Console
Corepack
Crypto
Debugger
Deprecated APIs
Diagnostics Channel
DNS
Domain
Node.js v18.16.0 | ▸ Table of contents | ▸ Index | ▸ Other versions | ▸ Options
fs.readFile(path[, options], callback)
▸ History
path <string> | <Buffer> | <URL> | <integer> filename or file descriptor
options <Object> | <string>
encoding <string> | <null> Default: null
flag <string> See support of file system flags. Default: 'r'.
signal <AbortSignal> allows aborting an in-progress readFile
callback <Function>
err <Error> | <AggregateError>
data <string> | <Buffer>
Asynchronously reads the entire contents of a file.
import { readFile } from 'node:fs';
readFile('/etc/passwd', (err, data) => {
if (err) throw err;
console.log(data);
});
The callback is passed two arguments (err, data), where data is the contents of the file.
If no encoding is specified, then the raw buffer is returned.
If options is a string, then it specifies the encoding:
import { readFile } from 'node:fs';


Backend > 2.2 Native Modules > JS index.js > ...
1 const fs = require("fs");
2
3 // fs.writeFile("message.txt", "Hello from NodeJS!", (err) => {
4 // if (err) throw err;
5 // console.log("The file has been saved!");
6 // });
7
8 fs.readFile("./message.txt", (err, data) => {
9 if (err) throw err;
10 console.log(data);
11 });
12


angelayu@Angelas-MacBook-Pro 2.2 Native Modules % node index.js
<Buffer 48 65 6c 6c 6f 20 66 72 6f 6d 20 4e 6f 64 65 4a 53 21 0a 48 65 6c 6c 6f 20 66 72 6f 6d 20 41 6e 67 65 6c 61 21>
angelayu@Angelas-MacBook-Pro 2.2 Native Modules %


Backend > 2.2 Native Modules > JS index.js > ...
1 const fs = require("fs");
2
3 // fs.writeFile("message.txt", "Hello from NodeJS!", (err) => {
4 //   if (err) throw err;
5 //   console.log("The file has been saved!");
6 // });
7
8 fs.readFile("./message.txt", "utf8", (err, data) => {
9   if (err) throw err;
10   console.log(data);
11 });
12

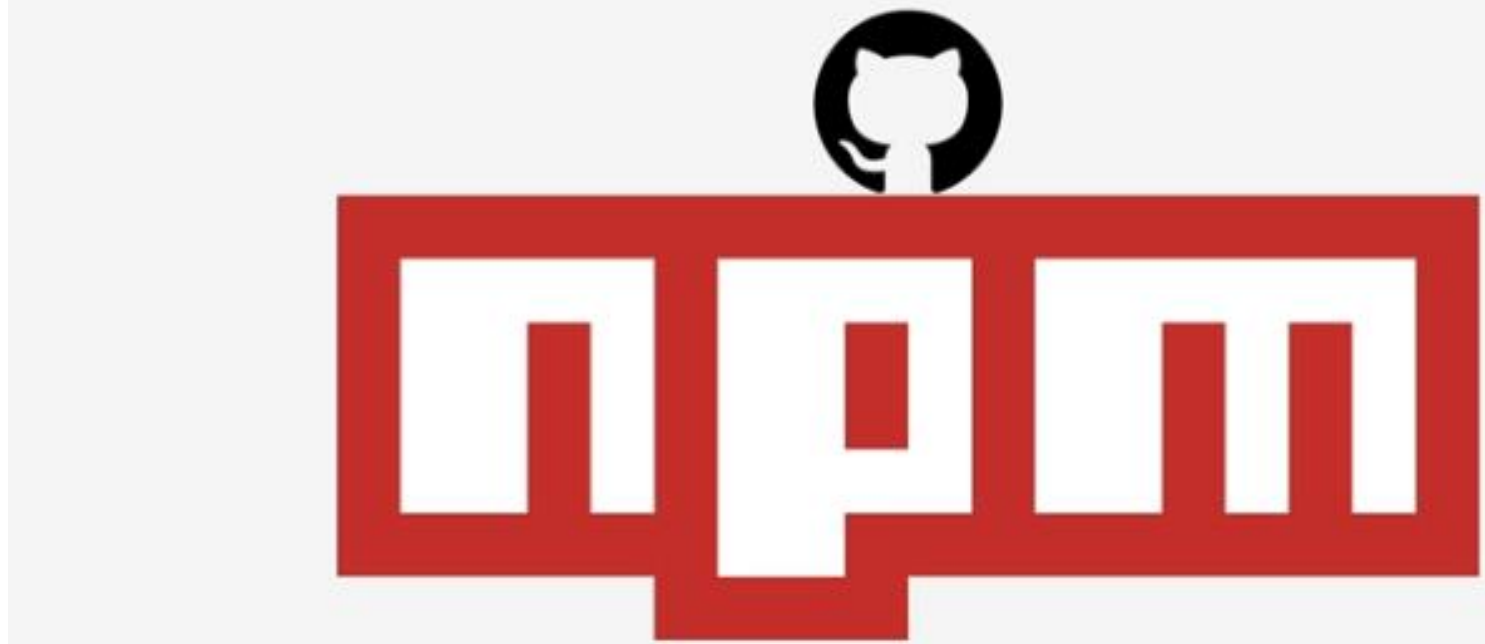

angelayu@Angelas-MacBook-Pro 2.2 Native Modules % node index.js
Hello from NodeJS!
Hello from Angela!

Node Package Manager

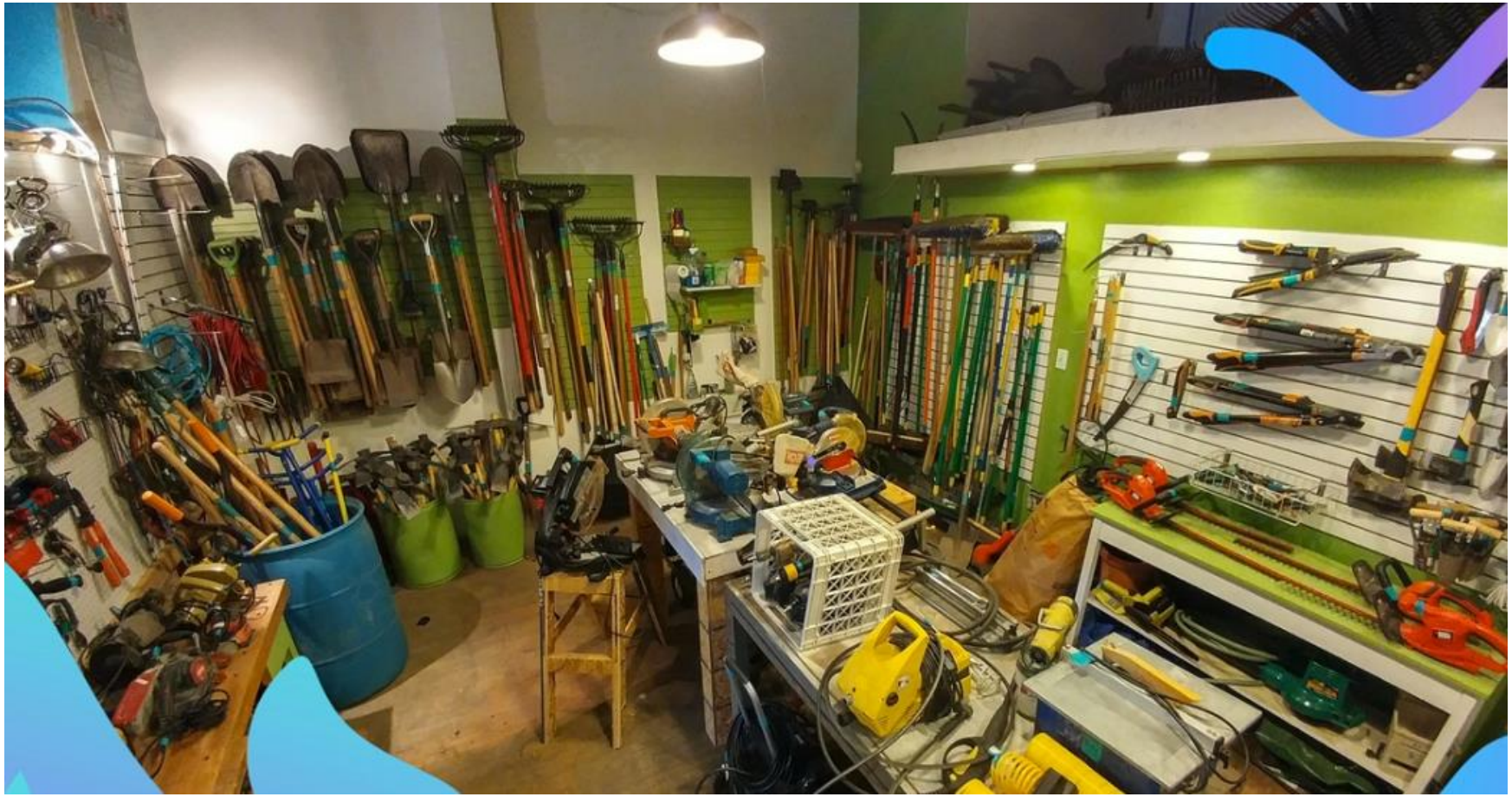
How to use NPM in your projects



Node Package Manager

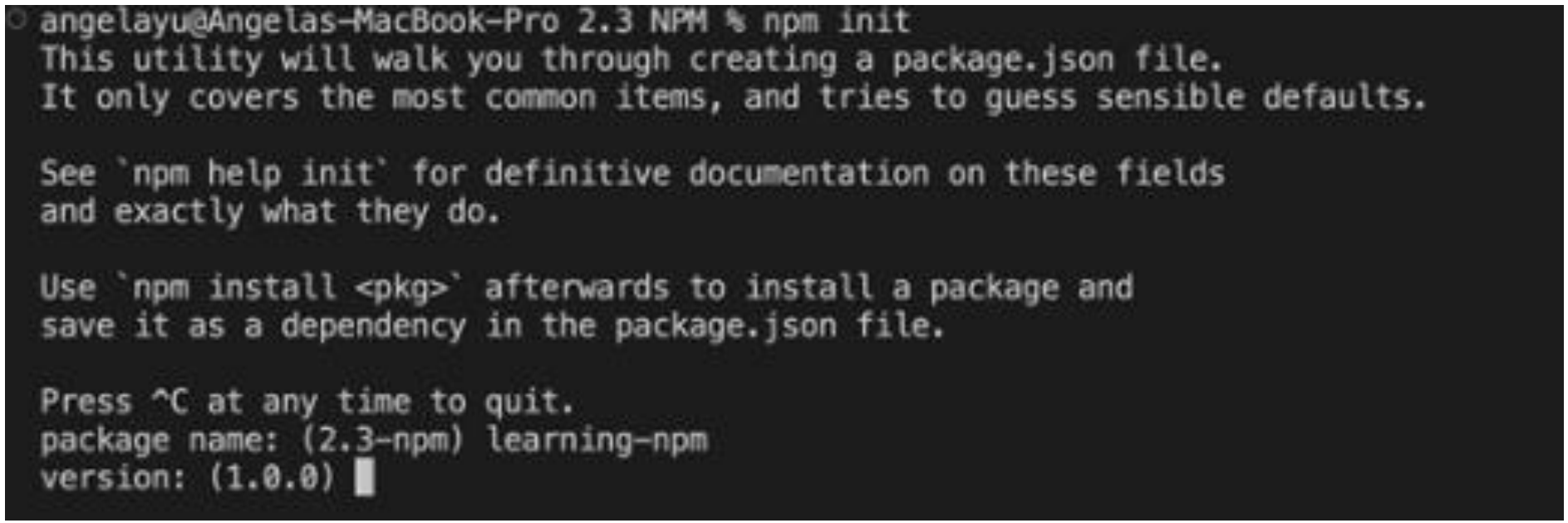


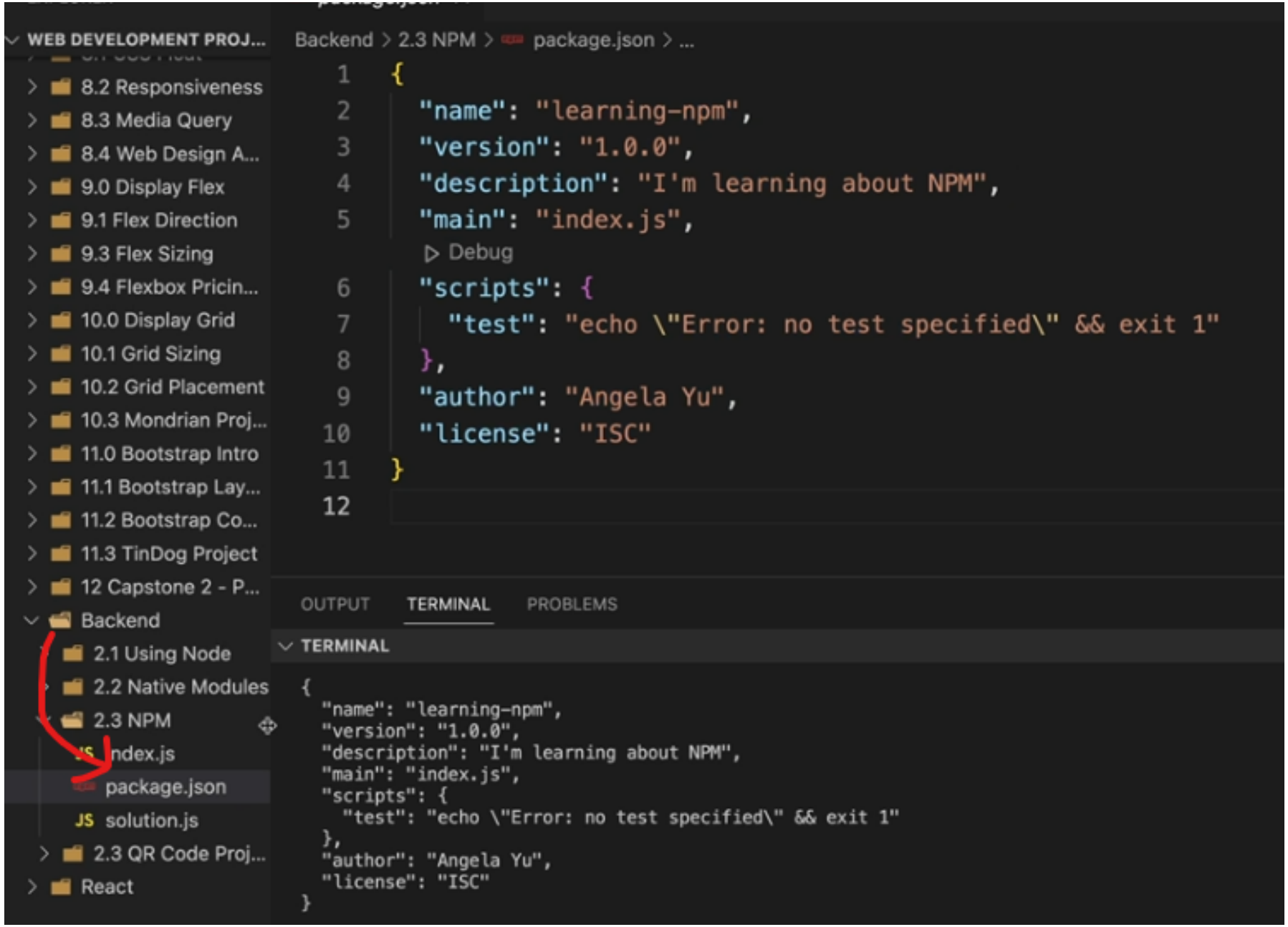
npm



Initialise NPM

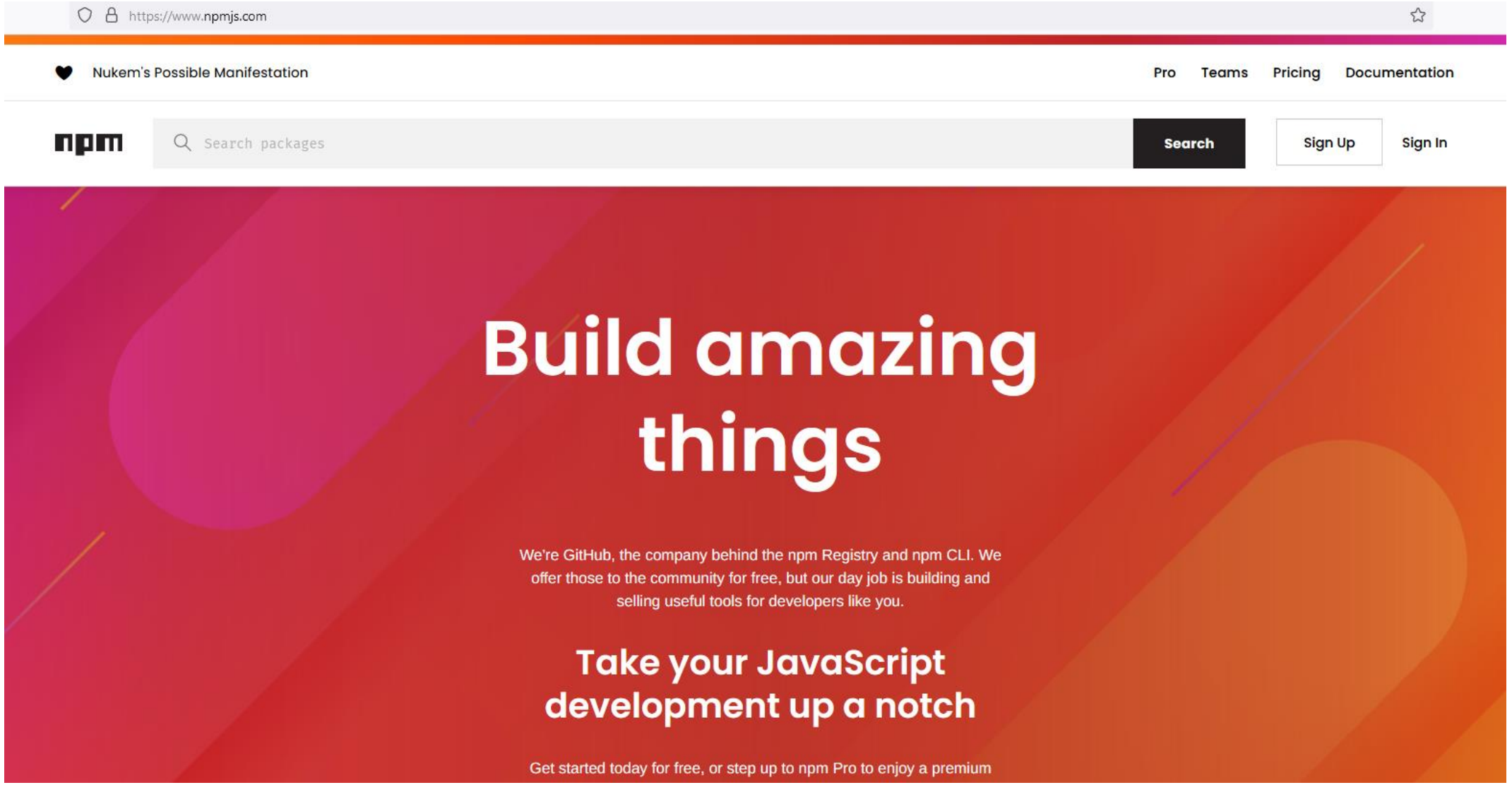
```
npm init
```


angelayu@Angelas-MacBook-Pro 2.3 NPM % npm init
This utility will walk you through creating a package.json file.
It only covers the most common items, and tries to guess sensible defaults.
See `npm help init` for definitive documentation on these fields
and exactly what they do.
Use `npm install <pkg>` afterwards to install a package and
save it as a dependency in the package.json file.
Press ^C at any time to quit.
package name: (2.3-npm) learning-npm
version: (1.0.0)


WEB DEVELOPMENT PROJ...
8.2 Responsiveness
8.3 Media Query
8.4 Web Design A...
9.0 Display Flex
9.1 Flex Direction
9.3 Flex Sizing
9.4 Flexbox Pricin...
10.0 Display Grid
10.1 Grid Sizing
10.2 Grid Placement
10.3 Mondrian Proj...
11.0 Bootstrap Intro
11.1 Bootstrap Lay...
11.2 Bootstrap Co...
11.3 TinDog Project
12 Capstone 2 - P...
Backend
2.1 Using Node
2.2 Native Modules
2.3 NPM
index.js
package.json
solution.js
2.3 QR Code Proj...
React
Backend > 2.3 NPM > package.json > ...
{
"name": "learning-npm",
"version": "1.0.0",
"description": "I'm learning about NPM",
"main": "index.js",
Debug
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
},
"author": "Angela Yu",
"license": "ISC"
}
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
{
"name": "learning-npm",
"version": "1.0.0",
"description": "I'm learning about NPM",
"main": "index.js",
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
},
"author": "Angela Yu",
"license": "ISC"
}

Install NPM Package

```
npm install <Something>
```


https://www.npmjs.com
Nukem's Possible Manifestation
Pro
Teams
Pricing
Documentation
npm
Search packages
Search
Sign Up
Sign In
Build amazing things
We're GitHub, the company behind the npm Registry and npm CLI. We offer those to the community for free, but our day job is building and selling useful tools for developers like you.
Take your JavaScript development up a notch
Get started today for free, or step up to npm Pro to enjoy a premium

♥ Never Prank Me

npm

sillyname

2 packages found

Sort Packages

- Optimal
- Popularity
- Quality
- Maintenance

sillyname exact match

A silly random name generator

node silly name generator random

thedeveloper published 0.1.0 • 8 years ago

happn-sillyname

A silly random name generator

node silly name generator random

smc-tenacious published 0.1.1 • 5 years ago

sillyname

0.1.0 • Public • Published 8 years ago

Readme | Code Beta | 0 Dependencies | 97 Dependents | 4 Versions

Silly name

A random name generator.

```
var generateName = require('sillyname');
var sillyName = generateName();
```

Some example results:

- Saltfollower Pickle
- Roadtoe Ginger
- Tidegargoyle Warp
- Geodecollar Daffodil
- Timeelk Fluff
- Slimetooth Black
- Coffeeparrot Giant
- Clevermoth Power
- Pinkserpent Quasar
- Greatgorilla Sunset

Install

```
> npm i sillyname
```

Repository

github.com/thedeveloper/sillyname

Homepage

github.com/thedeveloper/sillyname#re...

Weekly Downloads

8,989



Version

0.1.0

License

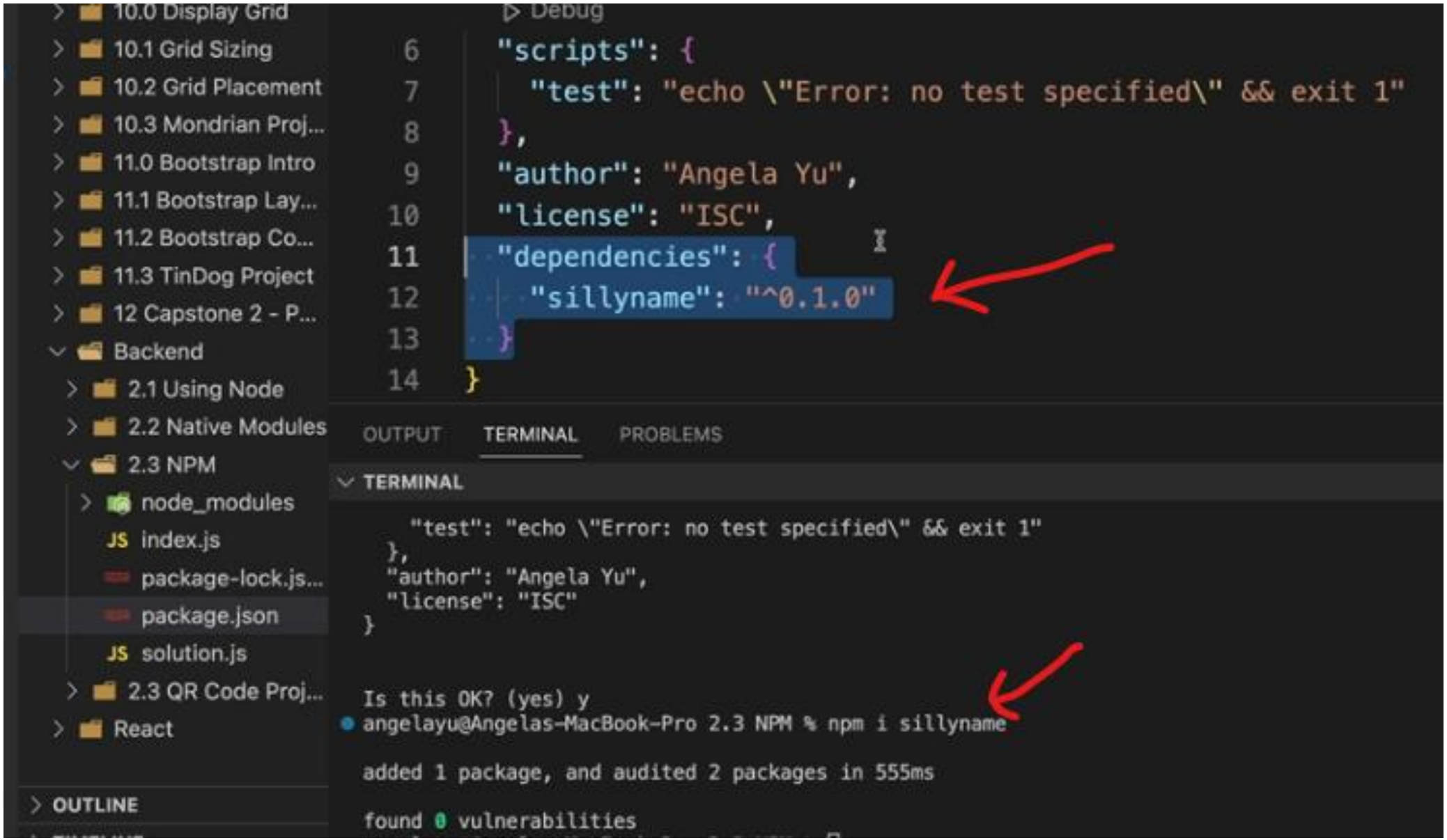
MIT

Issues

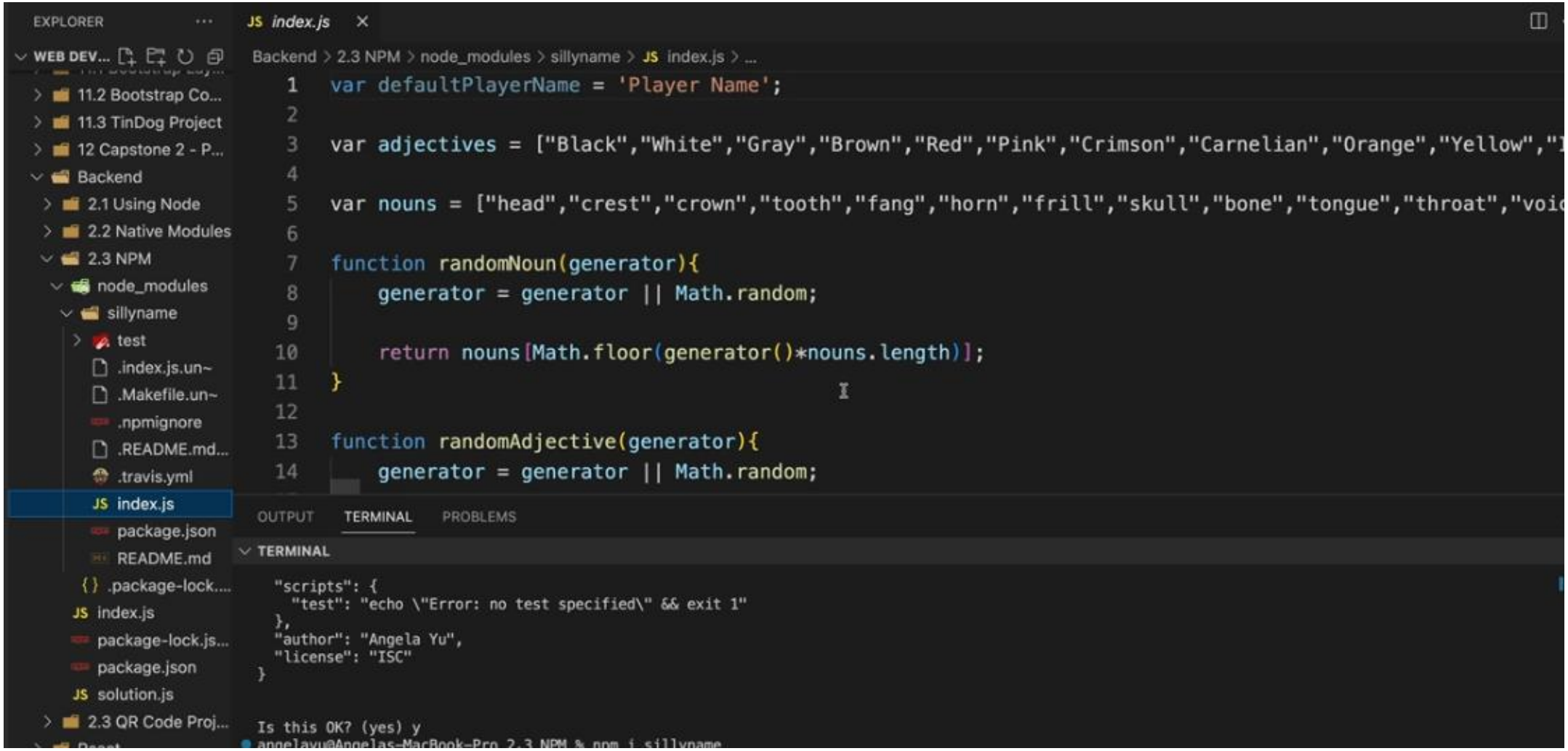
0

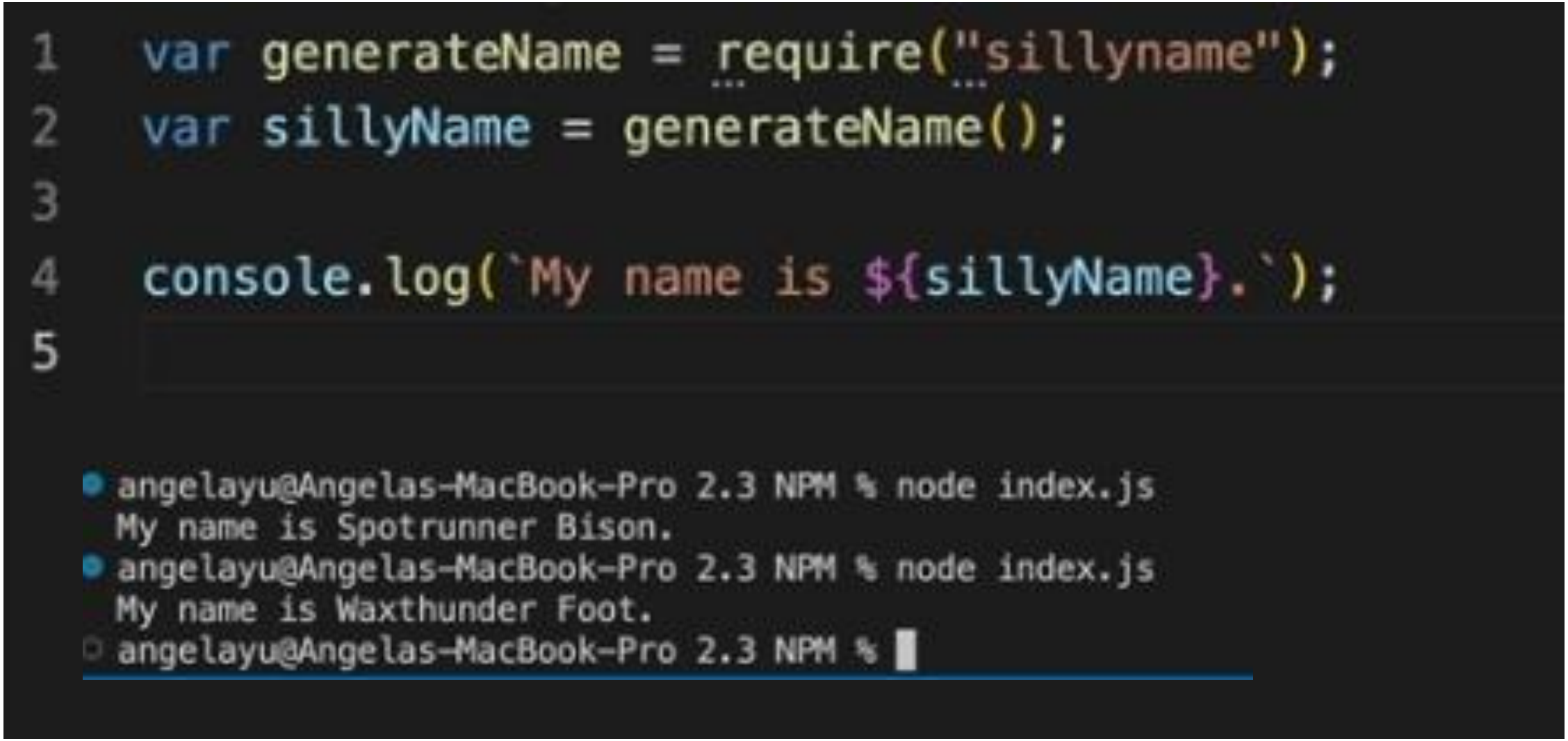
Pull Requests

2


10.0 Display Grid
10.1 Grid Sizing
10.2 Grid Placement
10.3 Mondrian Proj...
11.0 Bootstrap Intro
11.1 Bootstrap Lay...
11.2 Bootstrap Co...
11.3 TinDog Project
12 Capstone 2 - P...
Backend
2.1 Using Node
2.2 Native Modules
2.3 NPM
node_modules
index.js
package-lock.js...
package.json
solution.js
2.3 QR Code Proj...
React
OUTLINE
Debug
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
},
"author": "Angela Yu",
"license": "ISC",
"dependencies": {
"sillyname": "^0.1.0"
}
}
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
"test": "echo \"Error: no test specified\" && exit 1"
},
"author": "Angela Yu",
"license": "ISC"
}
Is this OK? (yes) y
angelayu@Angelas-MacBook-Pro 2.3 NPM % npm i sillyname
added 1 package, and audited 2 packages in 555ms
found 0 vulnerabilities

Sillyname folder




1 var generateName = require("sillyname");
2 var sillyName = generateName();
3
4 console.log(`My name is ${sillyName}.`);
5
angelayu@Angelas-MacBook-Pro 2.3 NPM % node index.js
My name is Spotrunner Bison.
angelayu@Angelas-MacBook-Pro 2.3 NPM % node index.js
My name is Waxthunder Foot.
angelayu@Angelas-MacBook-Pro 2.3 NPM %

ECMAScript Modules



de.js
t this documentation
e and example
rtion testing
chronous context
ing
c hooks
er
addons
+ addons with Node-
embedder API
l processes
ter
mand-line options
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pack
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ecated APIs
nostics Channel
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File system

Stability: 2 - Stable

Source Code: lib/fs.js

The `node:fs` module enables interacting with the file system in a way modeled on standard POSIX functions.

To use the promise-based APIs:

```
import * as fs from 'node:fs/promises';
```

CJS ESM

To use the callback and sync APIs:

```
import * as fs from 'node:fs';
```

CJS ESM

All file system operations have synchronous, callback, and promise-based forms, and are accessible using both CommonJS syntax and ES6 Modules (ESM).

Promise example

Promise-based operations return a promise that is fulfilled when the asynchronous operation is complete.

```
import { unlink } from 'node:fs/promises';

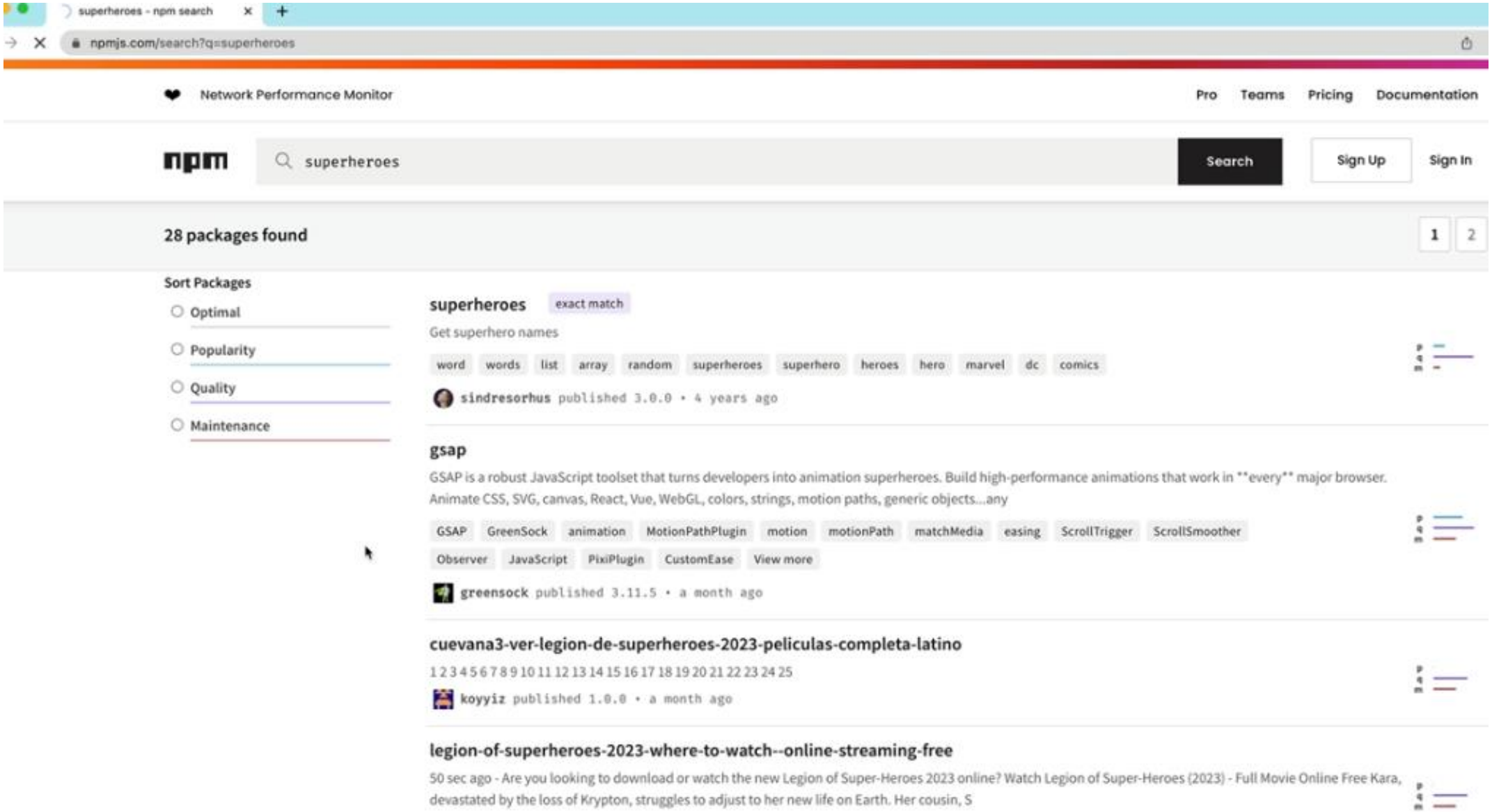
try {
  await unlink('/tmp/hello');
  console.log('successfully deleted /tmp/hello');
} catch (error) {
  console.error('there was an error:', error.message);
}
```

CJS ESM


JS index.js
package.json
Backend > 2.3 NPM > package.json > type
{
"name": "learning-npm",
"version": "1.0.0",
"description": "I'm learning about NPM",
"main": "index.js",
"type": "module",
Debug
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
},
"author": "Angela Yu",
"license": "ISC",
"dependencies": {
"sillyname": "^0.1.0"
}
}


JS index.js
Backend > 2.3 NPM > JS index.js > ...
1 // var generateName = require("sillyname");
2
3 import generateName from "sillyname";
4 var sillyName = generateName();
5
6 console.log(`My name is ${sillyName}.`);
7



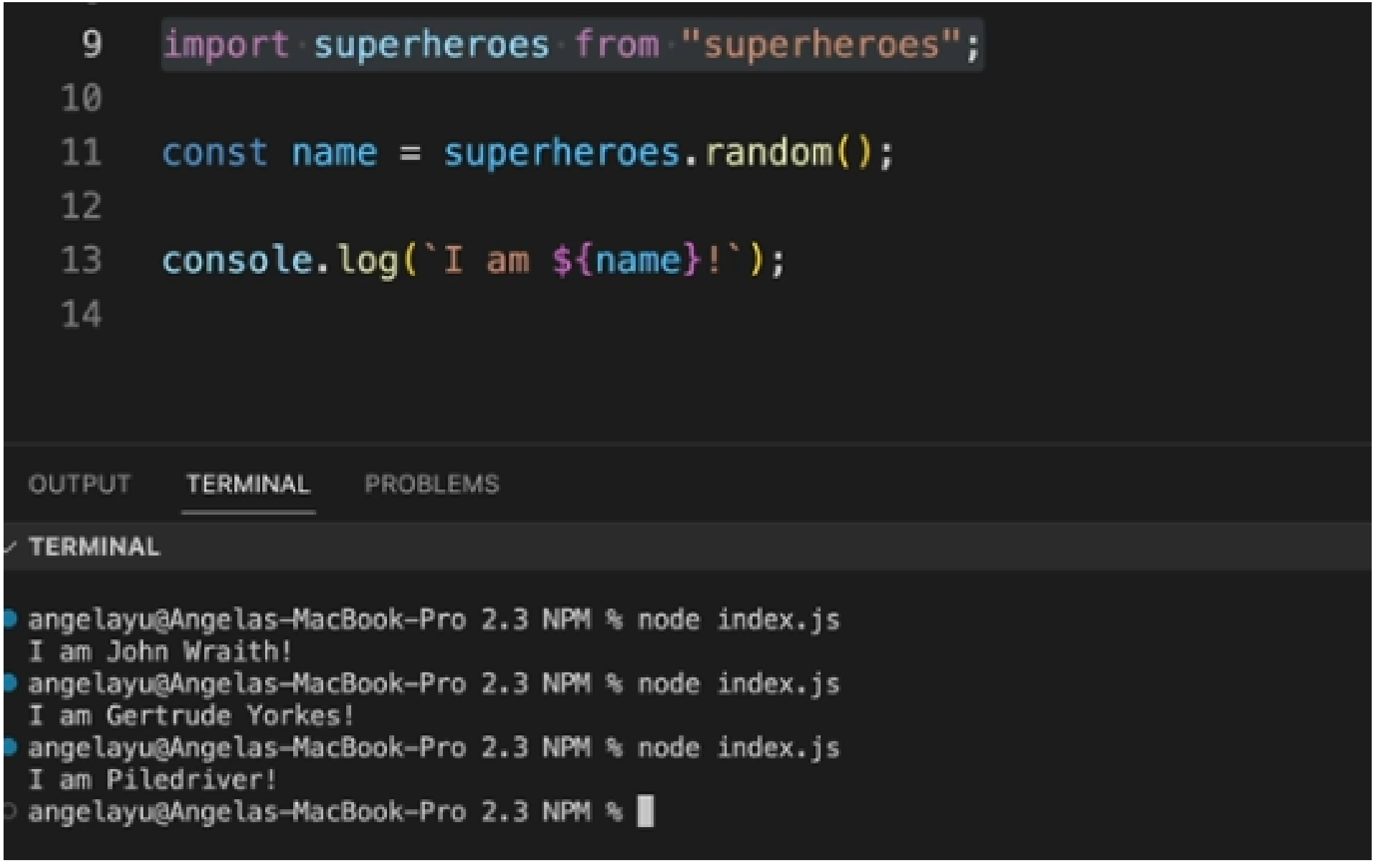

superheroes - npm search
npmjs.com/search?q=superheroes
Network Performance Monitor
Pro
Teams
Pricing
Documentation
npm
superheroes
Search
Sign Up
Sign In
28 packages found
1
2
Sort Packages
Optimal
Popularity
Quality
Maintenance
superheroes
exact match
Get superhero names
word
words
list
array
random
superheroes
superhero
heroes
hero
marvel
dc
comics
sindresorhus published 3.0.0 • 4 years ago
gsap
GSAP is a robust JavaScript toolset that turns developers into animation superheroes. Build high-performance animations that work in **every** major browser. Animate CSS, SVG, canvas, React, Vue, WebGL, colors, strings, motion paths, generic objects...any
GSAP
GreenSock
animation
MotionPathPlugin
motion
motionPath
matchMedia
easing
ScrollTrigger
ScrollSmoother
Observer
JavaScript
PixiPlugin
CustomEase
View more
greensock published 3.11.5 • a month ago
cuevana3-ver-legion-de-superheroes-2023-peliculas-completa-latino
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
koyyiz published 1.0.0 • a month ago
legion-of-superheroes-2023-where-to-watch--online-streaming-free
50 sec ago - Are you looking to download or watch the new Legion of Super-Heroes 2023 online? Watch Legion of Super-Heroes (2023) - Full Movie Online Free Kara, devastated by the loss of Krypton, struggles to adjust to her new life on Earth. Her cousin, S

- npm i superheroes

```
angelayu@Angelas-MacBook-Pro 2.3 NPM % npm i superheroes

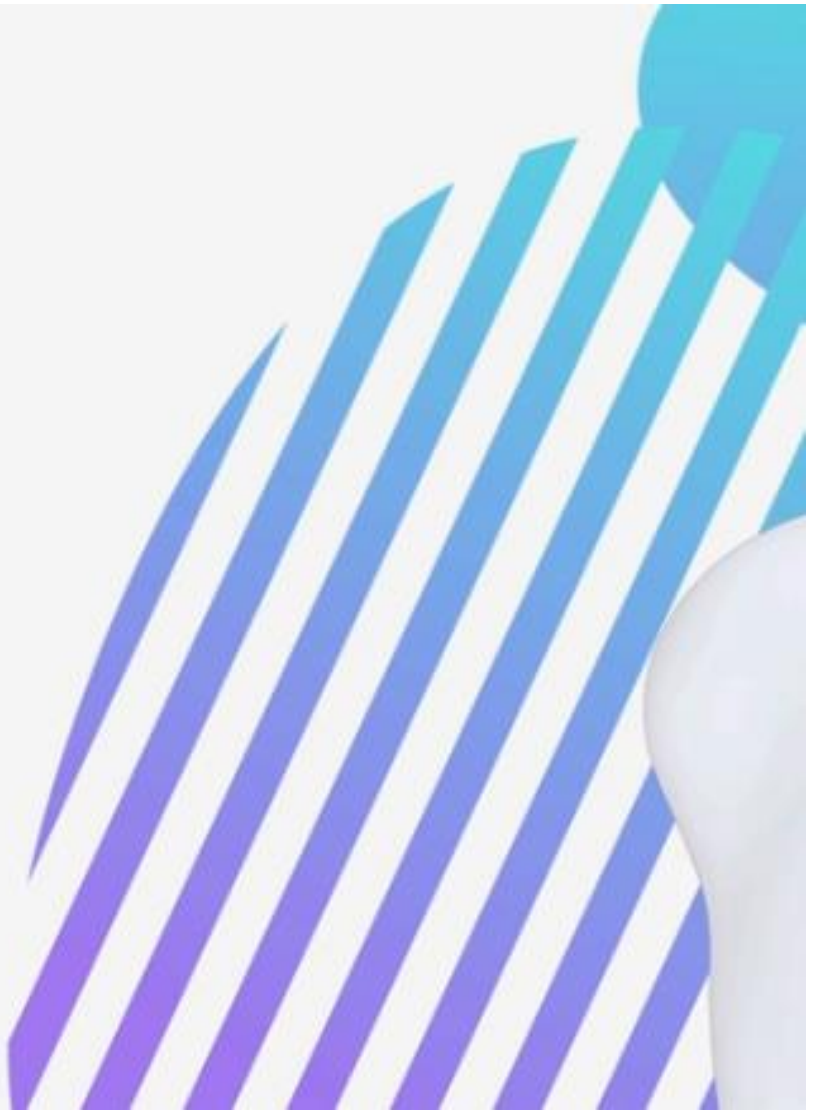
added 3 packages, and audited 5 packages in 2s

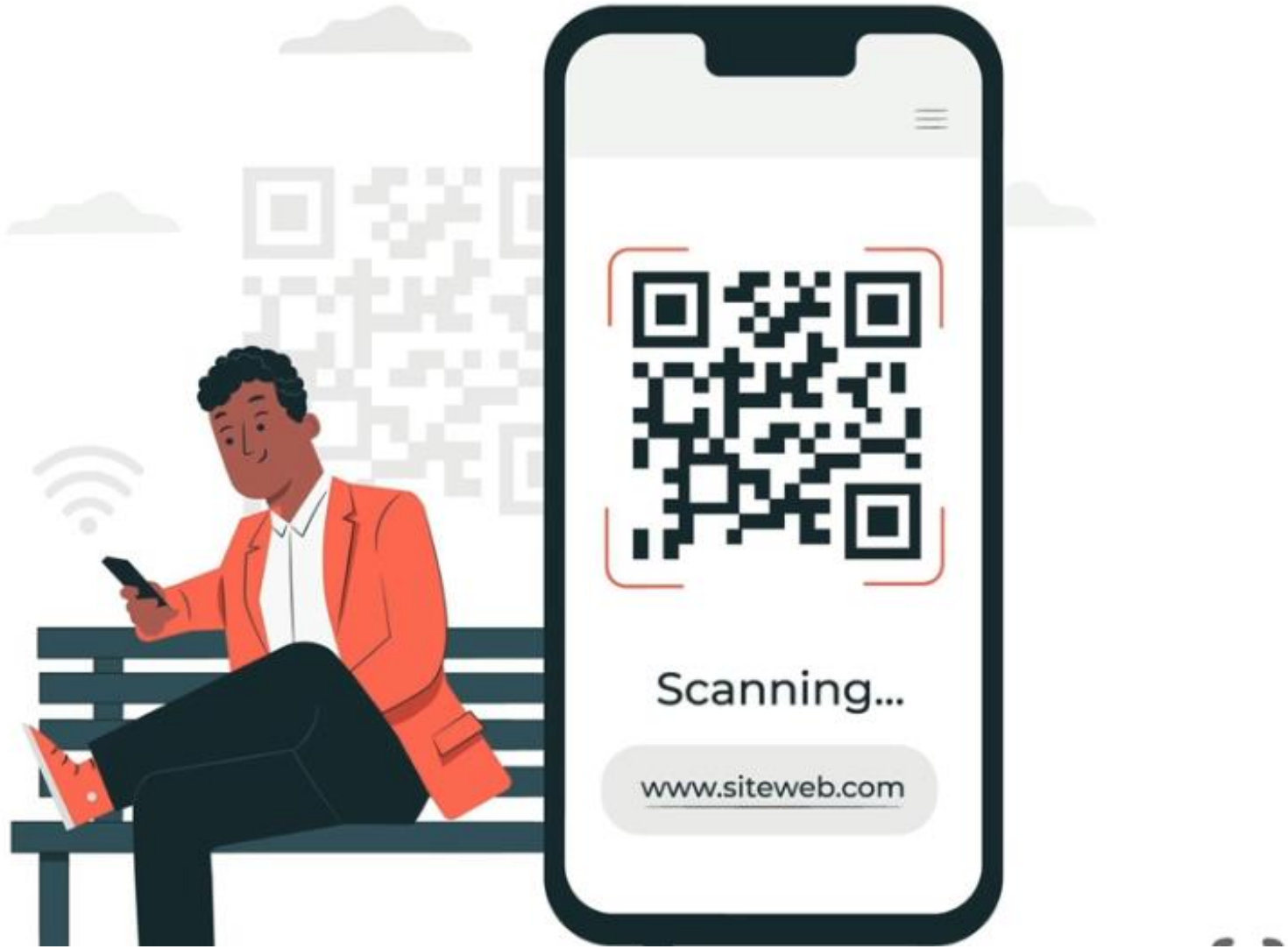
found 0 vulnerabilities
```

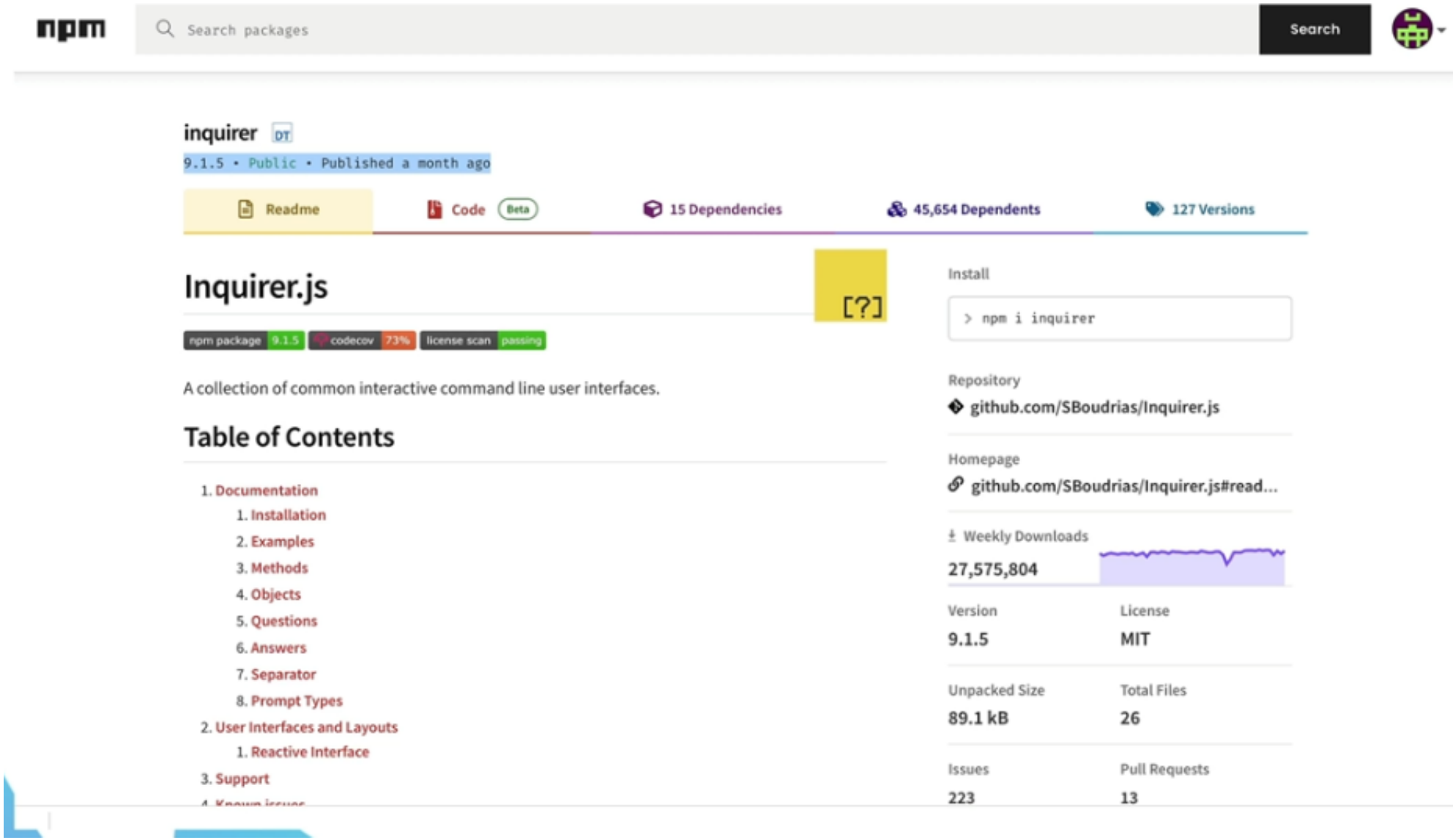

9 import superheroes from "superheroes";
10
11 const name = superheroes.random();
12
13 console.log(`I am ${name}!`);
14
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
angelayu@Angelas-MacBook-Pro 2.3 NPM % node index.js
I am John Wraith!
angelayu@Angelas-MacBook-Pro 2.3 NPM % node index.js
I am Gertrude Yorkes!
angelayu@Angelas-MacBook-Pro 2.3 NPM % node index.js
I am Piledriver!
angelayu@Angelas-MacBook-Pro 2.3 NPM %

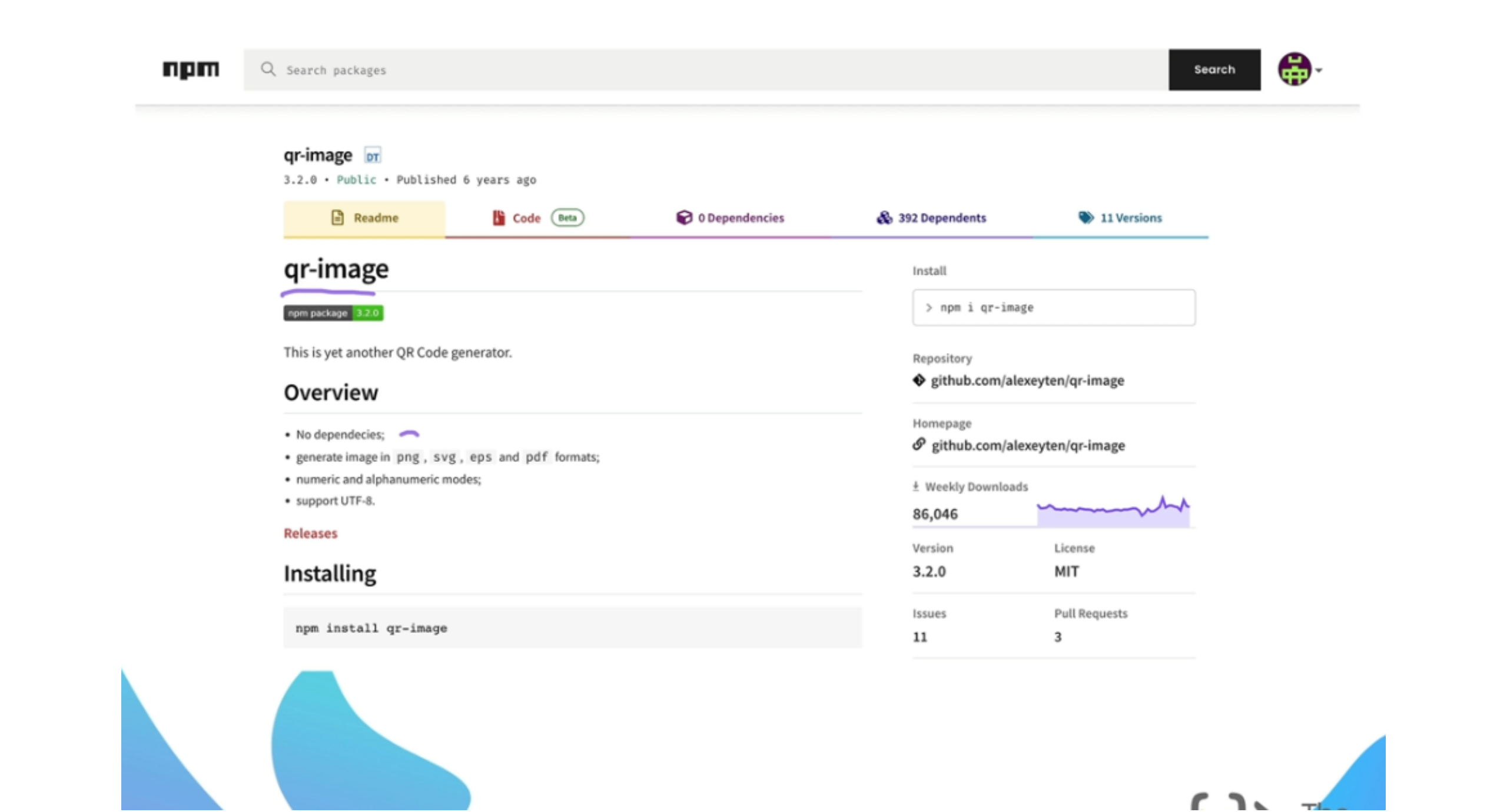
QR Code Generator

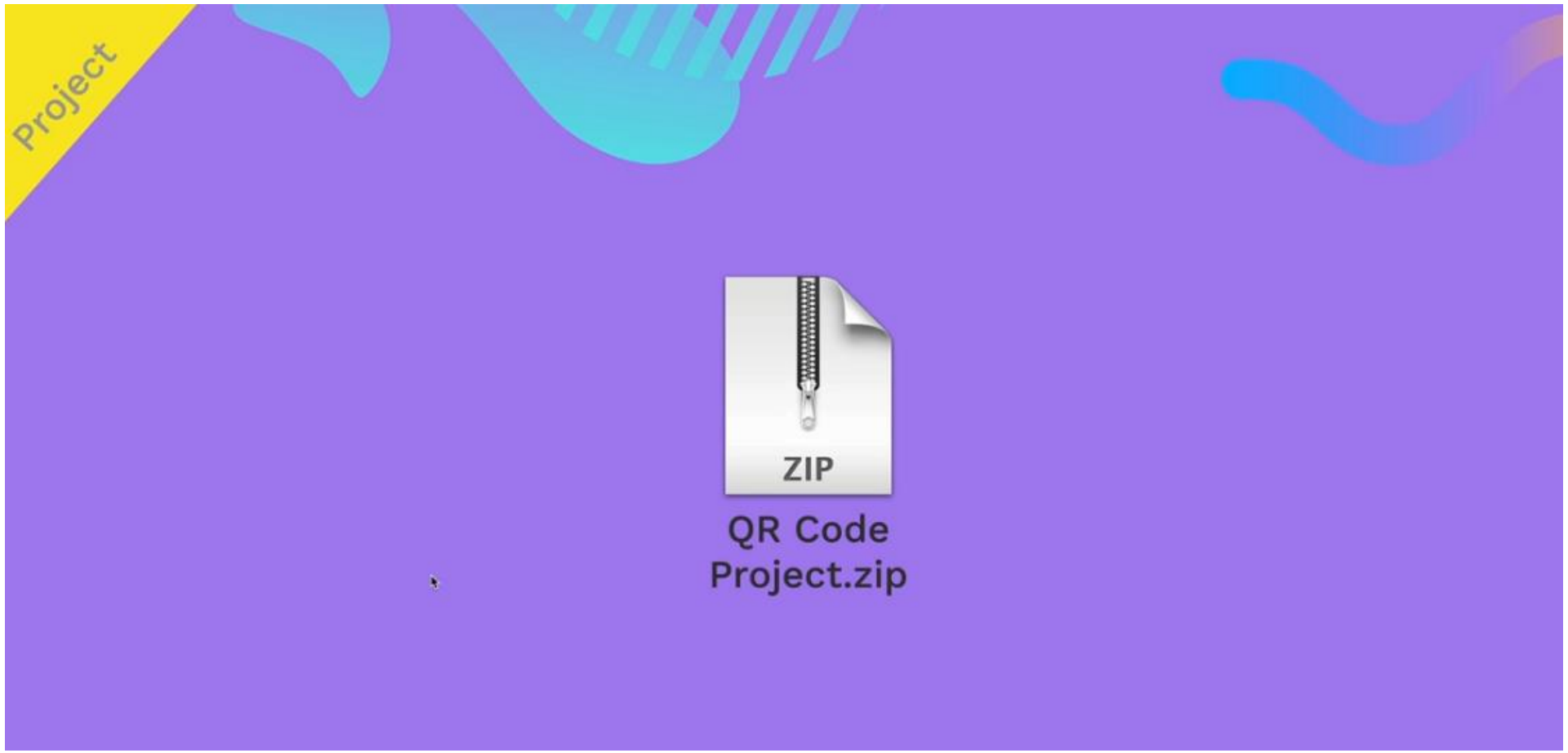
Using Node to generate QR codes for URLs

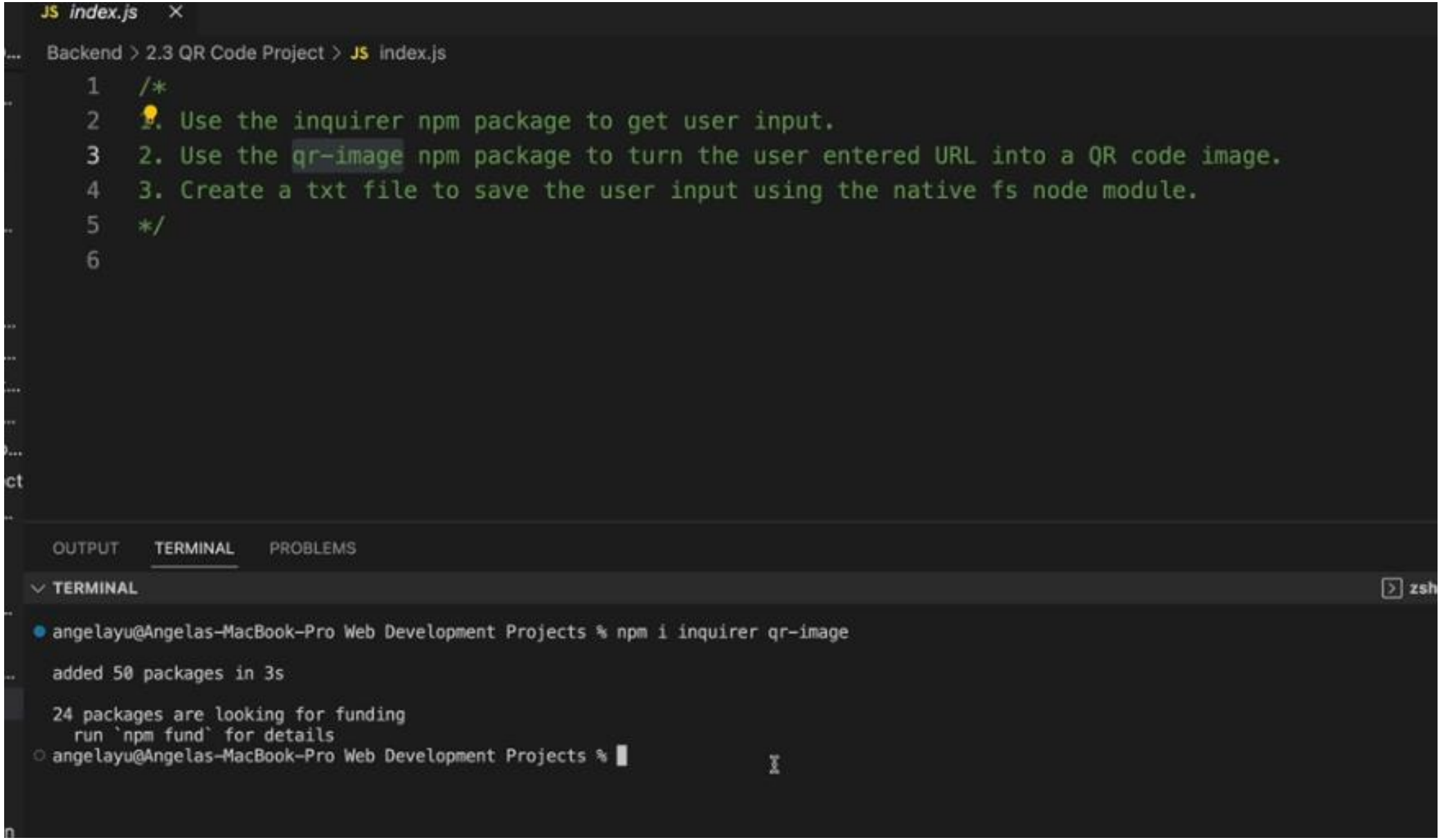


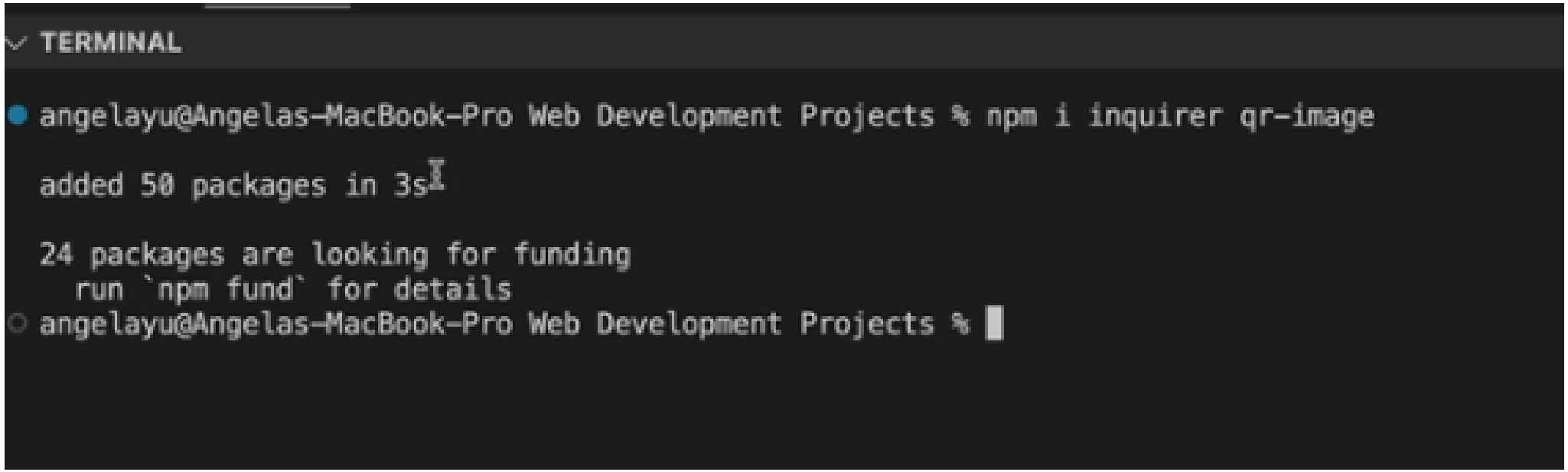

Scanning...
www.siteweb.com

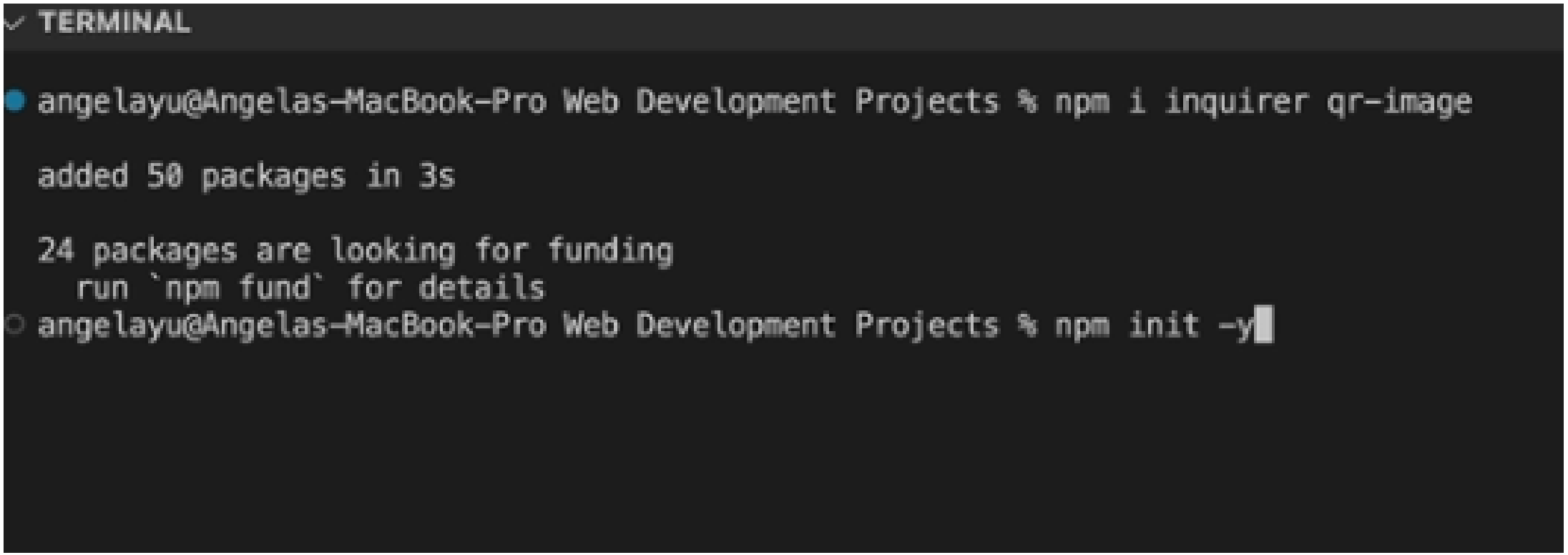

npm
Search packages
Search
inquirer
DT
9.1.5 • Public • Published a month ago
Readme
Code
Beta
15 Dependencies
45,654 Dependents
127 Versions
Inquirer.js
[?]
npm package 9.1.5
codecov 73%
license scan passing
A collection of common interactive command line user interfaces.
Table of Contents
1. Documentation
1. Installation
2. Examples
3. Methods
4. Objects
5. Questions
6. Answers
7. Separator
8. Prompt Types
2. User Interfaces and Layouts
1. Reactive Interface
3. Support
Install
> npm i inquirer
Repository
github.com/SBoudrias/Inquirer.js
Homepage
github.com/SBoudrias/Inquirer.js#read...
Weekly Downloads
27,575,804
Version
9.1.5
License
MIT
Unpacked Size
89.1 kB
Total Files
26
Issues
223
Pull Requests
13

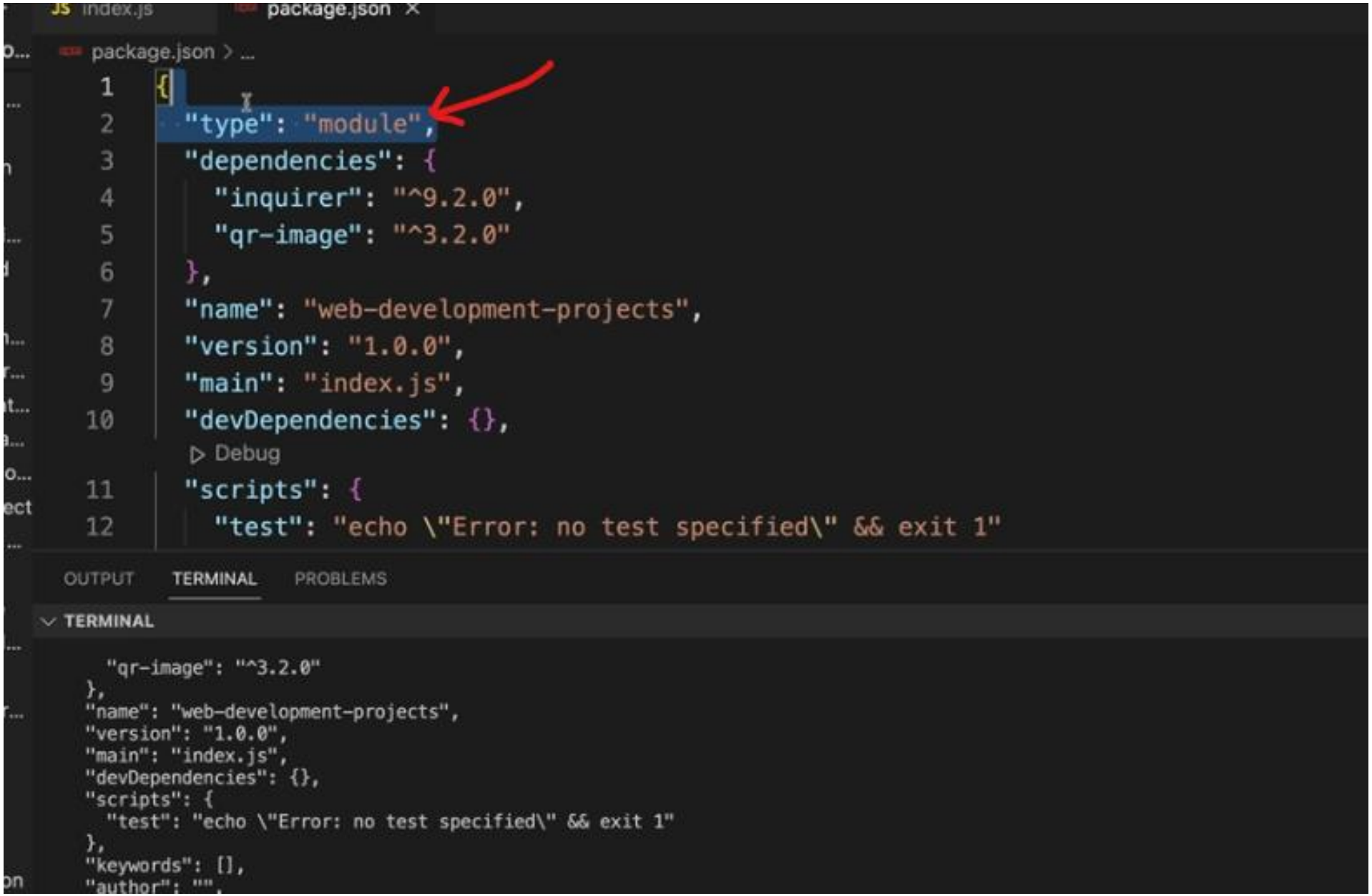

npm
Search packages
Search
qr-image
DT
3.2.0 • Public • Published 6 years ago
Readme
Code
Beta
0 Dependencies
392 Dependents
11 Versions
qr-image
npm package 3.2.0
This is yet another QR Code generator.
Overview
• No dependecies;
• generate image in png , svg , eps and pdf formats;
• numeric and alphanumeric modes;
• support UTF-8.
Releases
Installing
npm install qr-image
Install
> npm i qr-image
Repository
github.com/alexeyten/qr-image
Homepage
github.com/alexeyten/qr-image
Weekly Downloads
86,046
Version
3.2.0
License
MIT
Issues
11
Pull Requests
3


Project
ZIP
QR Code
Project.zip


JS index.js
Backend > 2.3 QR Code Project > JS index.js
1 /*
2 1. Use the inquirer npm package to get user input.
3 2. Use the qr-image npm package to turn the user entered URL into a QR code image.
4 3. Create a txt file to save the user input using the native fs node module.
5 */
6
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
zsh
angelayu@Angelas-MacBook-Pro Web Development Projects % npm i inquirer qr-image
added 50 packages in 3s
24 packages are looking for funding
run `npm fund` for details
angelayu@Angelas-MacBook-Pro Web Development Projects %


TERMINAL
angelayu@Angelas-MacBook-Pro Web Development Projects % npm i inquirer qr-image
added 50 packages in 3s
24 packages are looking for funding
run `npm fund` for details
angelayu@Angelas-MacBook-Pro Web Development Projects %


TERMINAL
angelayu@Angelas-MacBook-Pro Web Development Projects % npm i inquirer qr-image
added 50 packages in 3s
24 packages are looking for funding
run `npm fund` for details
angelayu@Angelas-MacBook-Pro Web Development Projects % npm init -y


index.js
package.json
package.json > ...
{
"type": "module",
"dependencies": {
"inquirer": "^9.2.0",
"qr-image": "^3.2.0"
},
"name": "web-development-projects",
"version": "1.0.0",
"main": "index.js",
"devDependencies": {},
Debug
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
"qr-image": "^3.2.0"
},
"name": "web-development-projects",
"version": "1.0.0",
"main": "index.js",
"devDependencies": {},
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
},
"keywords": [],
"author": "",

> **Note:** `Inquirer.js` provides the user interface and the inquiry session flow. If you're searching for a full blown command line program utility, then check out commander, vorpal or args.

Documentation

Installation

```
npm install --save inquirer
```

```
import inquirer from 'inquirer';

inquirer
  .prompt([
    /* Pass your questions in here */
  ])
  .then((answers) => {
    // Use user feedback for... whatever!!
  })
  .catch((error) => {
    if (error.isTtyError) {
      // Prompt couldn't be rendered in the current environment
    } else {
      // Something else went wrong
    }
  });
```

Backend > 2.3 QR Code Project > JS index.js > ...

```
1  import inquirer from "inquirer";
2
3  inquirer
4    .prompt([
5      /* Pass your questions in here */
6    ])
7    .then((answers) => {
8      // Use user feedback for... whatever!!
9    })
10   .catch((error) => {
11     if (error.isTtyError) {
12       // Prompt couldn't be rendered in the current environment
13     } else {
14       // Something else went wrong
15     }
16   });
17 /*
18 1. Use the inquirer npm package to get user input.
19
20
```

`inquirer.registerPrompt(name, prompt)`

Register prompt plugins under `name`.

- **name** (string) name of the this new prompt. (used for question `type`)
- **prompt** (object) the prompt object itself (the plugin)

`inquirer.createPromptModule() -> prompt function`

Create a self contained inquirer module. If you don't want to affect other libraries that also rely on inquirer when you overwrite or add new prompt types.

```
const prompt = inquirer.createPromptModule();

prompt(questions).then(/* ... */);
```

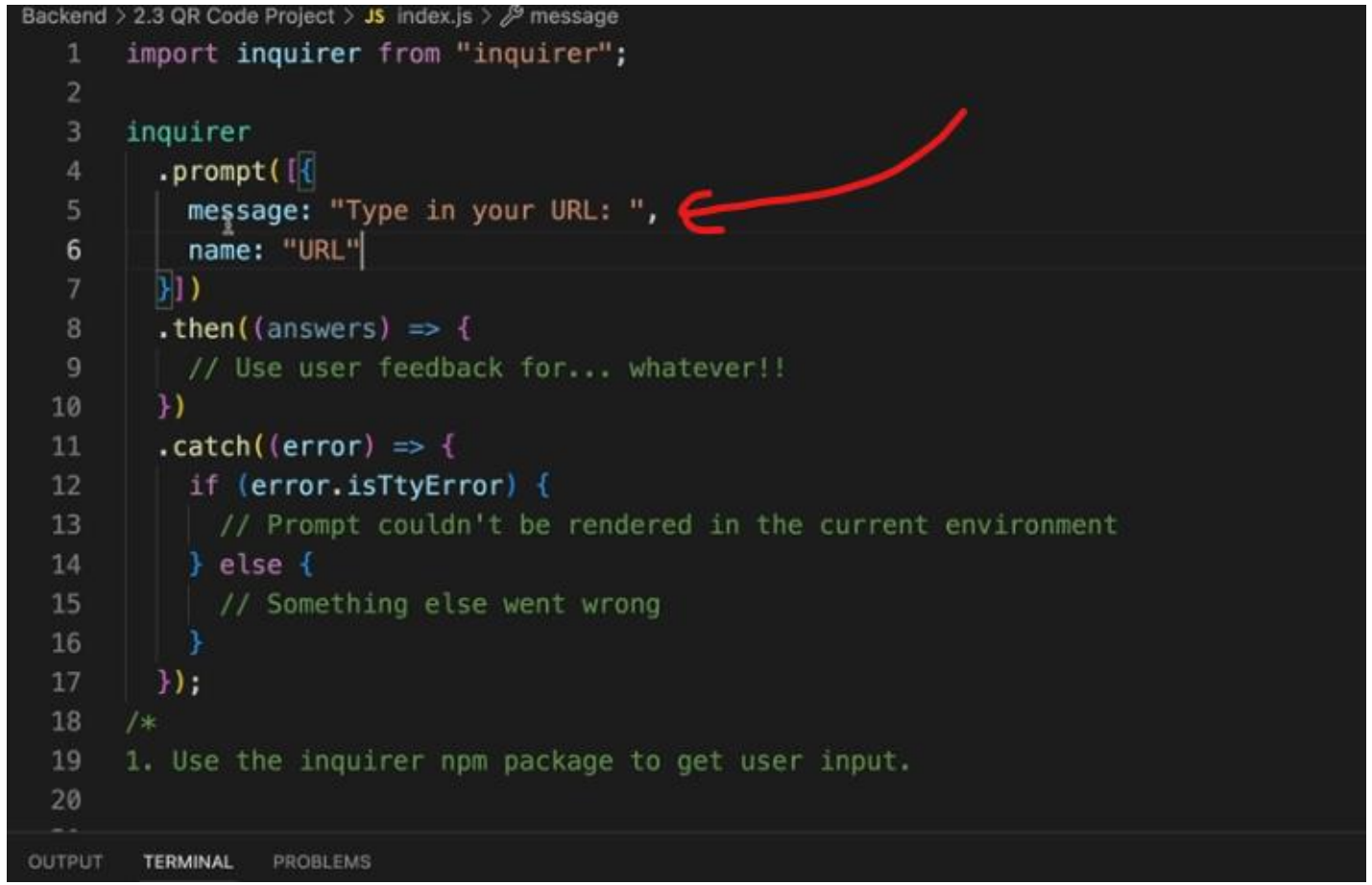


Objects

Question

A question object is a `hash` containing question related values:

- **type**: (String) Type of the prompt. Defaults: `input` - Possible values: `input`, `number`, `confirm`, `list`, `rawlist`, `expand`, `checkbox`, `password`, `editor`
- **name**: (String) The name to use when storing the answer in the answers hash. If the name contains periods, it will define a path in the answers hash.
- **message**: (String|Function) The question to print. If defined as a function, the first parameter will be the current inquirer session answers. Defaults to the value of `name` (followed by a colon).
- **default**: (String|Number|Boolean|Array|Function) Default value(s) to use if nothing is entered, or a function that returns the default value(s). If defined as a function, the first parameter will be the current inquirer session answers.
- **choices**: (Array|Function) Choices array or a function returning a choices array. If defined as a function, the first parameter will be the current inquirer session answers. Array values can be simple `numbers`, `strings`, or `objects` containing a `name` (to display in list), a `value` (to save in the answers hash), and a `short` (to display after selection) properties. The choices array can also contain a `Separator`.
- **validate**: (Function) Receive the user input and answers hash. Should return `true` if the value is valid, and an error message (`String`) otherwise. If `false` is returned, a default error message is provided.


Backend > 2.3 QR Code Project > JS index.js > message
import inquirer from "inquirer";
inquirer
.prompt([{
message: "Type in your URL: ",
name: "URL"
}])
.then((answers) => {
// Use user feedback for... whatever!!
})
.catch((error) => {
if (error.isTtyError) {
// Prompt couldn't be rendered in the current environment
} else {
// Something else went wrong
}
});
/*
1. Use the inquirer npm package to get user input.
OUTPUT
TERMINAL
PROBLEMS

Backend > 2.3 QR Code Project > JS index.js > then() callback

```
1  import inquirer from "inquirer";
2
3  inquirer
4    .prompt([
5      {
6        message: "Type in your URL: ",
7        name: "URL",
8      },
9    ])
10   .then((answers) => {
11     console.log(answers);
12   })
13   .catch((error) => {
14     if (error.isTtyError) {
15       // Prompt couldn't be rendered in the current environment
16     } else {
17       // Something else went wrong
18     }
```

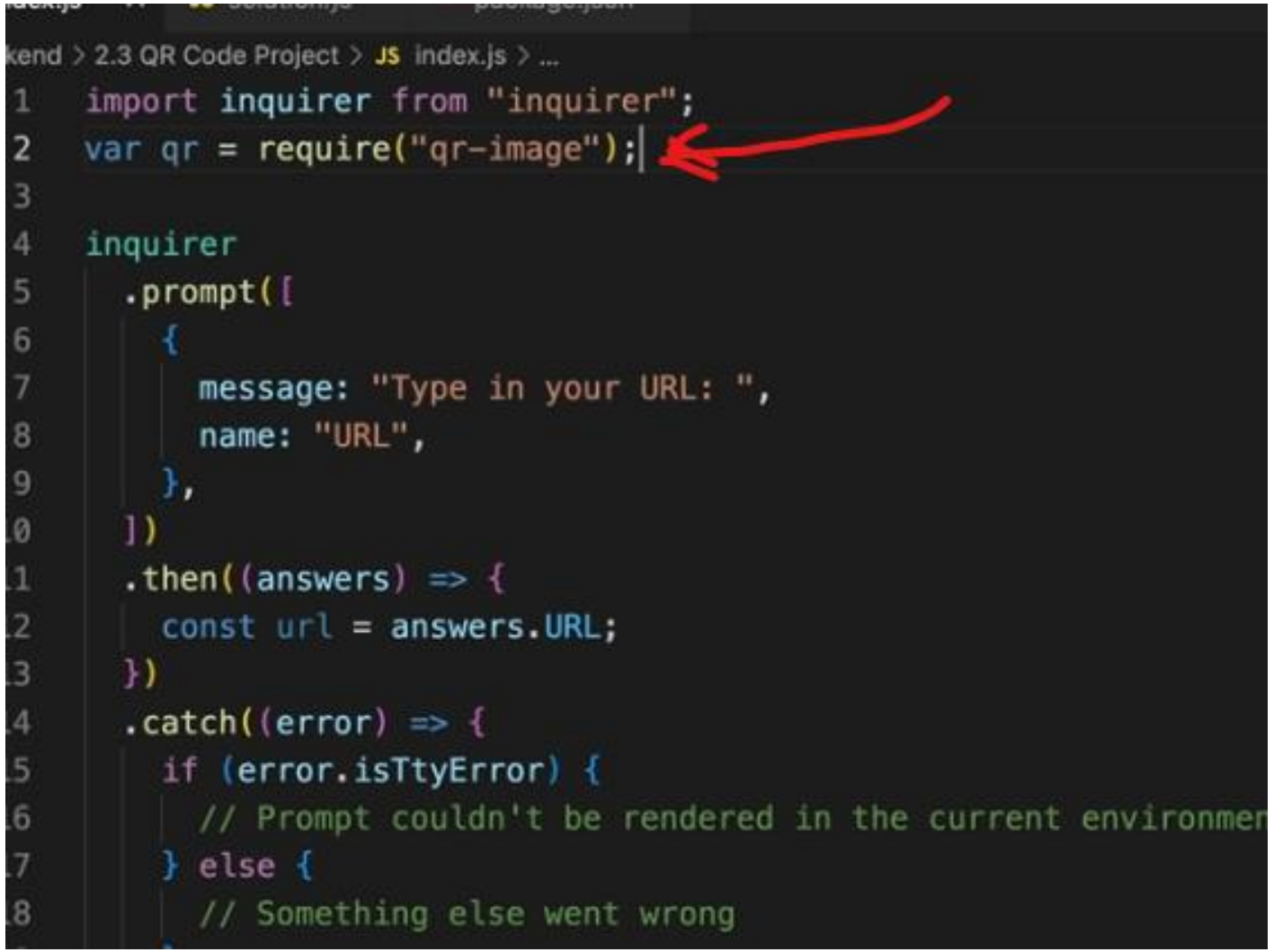
OUTPUT TERMINAL PROBLEMS

TERMINAL

```
angelayu@Angelas-MacBook-Pro Web Development Projects % cd '/Users/angelayu/Web Development Pro
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project % node index.js
? Type in your URL:  www.google.com
{ URL: 'www.google.com' }
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project %
```

```
.then((answers) => {
  const url = answers.URL;
})
```

Qr-image


kend > 2.3 QR Code Project > JS index.js > ...
import inquirer from "inquirer";
var qr = require("qr-image");
inquirer
.prompt([
{
message: "Type in your URL: ",
name: "URL",
},
])
.then((answers) => {
const url = answers.URL;
})
.catch((error) => {
if (error.isTtyError) {
// Prompt couldn't be rendered in the current environmer
} else {
// Something else went wrong

Usage

Example:

```
var qr = require('qr-image');

var qr_svg = qr.image('I love QR!', { type: 'svg' });
qr_svg.pipe(require('fs').createWriteStream('i_love_qr.svg'));

var svg_string = qr.imageSync('I love QR!', { type: 'svg' });
```

More examples

```
qr = require('qr-image')
```

Methods

- `qr.image(text, [ec_level | options])` — Readable stream with image data;
- `qr.imageSync(text, [ec_level | options])` — string with image data. (Buffer for `png`);
- `qr.svgObject(text, [ec_level | options])` — object with SVG path and size;
- `qr.matrix(text, [ec_level])` — 2D array.

Options

- `text` — text to encode;
- `ec_level` — error correction level. One of `L`, `M`, `Q`, `H`. Default `M`.
- `options` — image options object:
  - `ec_level` — default `M`.
  - `type` — image type. Possible values `png` (default), `svg`, `pdf` and `eps`.
  - `size` (png and svg only) — size of one module in pixels. Default `5` for png and `undefined` for svg.
  - `margin` — white space around QR image in modules. Default `4` for `png` and `1` for others.
  - `customize` (only png) — function to customize qr bitmap before encoding to PNG.

Last publish
6 years ago
Collaborators
>_Try on R
Report m

kend > 2.3 QR Code Project > JS index.js > ...

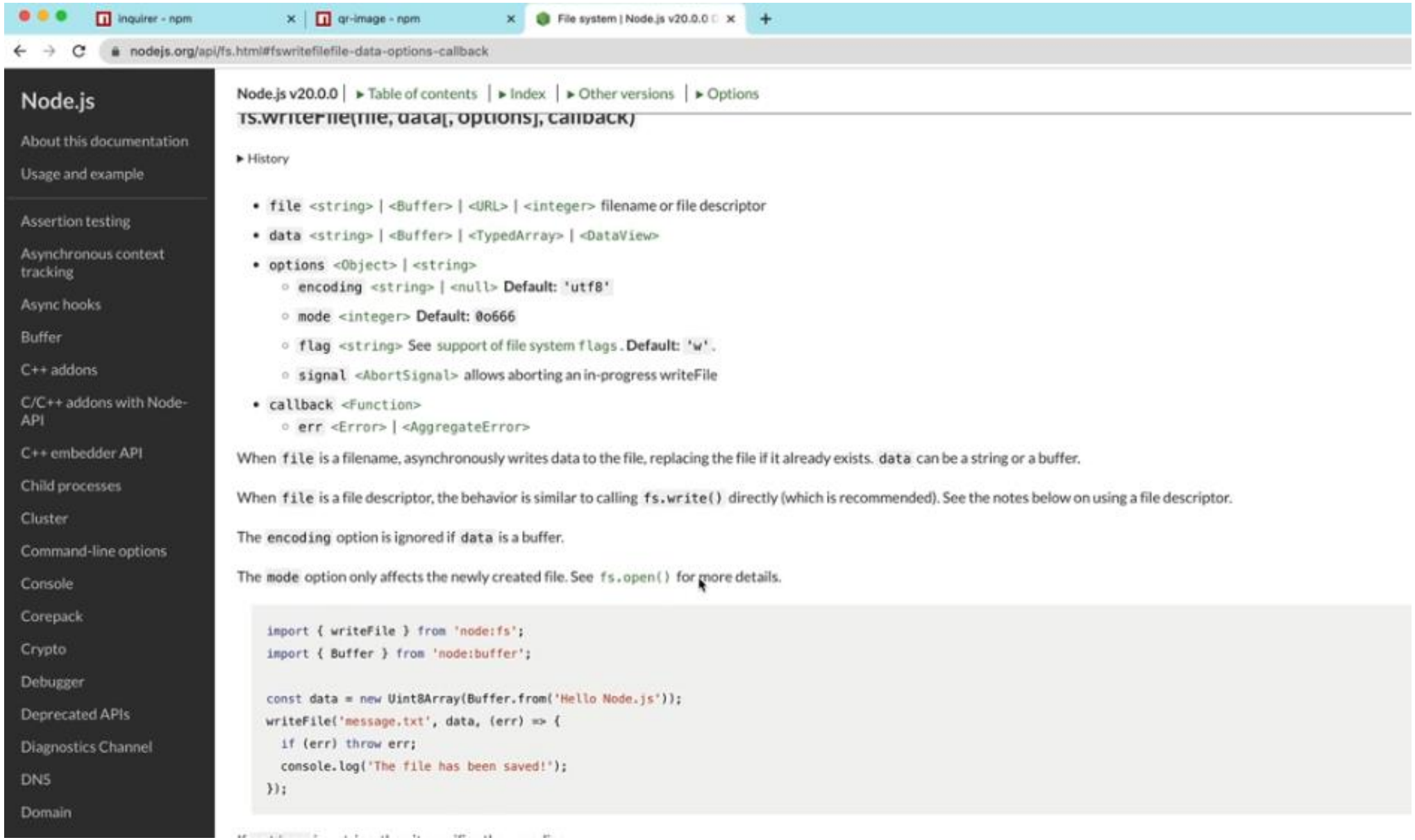
```
1  import inquirer from "inquirer";
2  import qr from "qr-image";
3
4  inquirer
5    .prompt([
6      {
7        message: "Type in your URL: ",
8        name: "URL",
9      },
10   ])
11   .then((answers) => {
12     const url = answers.URL;
13     var qr_svg = qr.image(url);
14     qr_svg.pipe(require("fs").createWriteStream("qr_img.png"));
15   })
16   .catch((error) => {
17     if (error.isTtyError) {
18       // Prompt couldn't be rendered in the current environment
19     } else {
20       // Something else went wrong
21     }
```

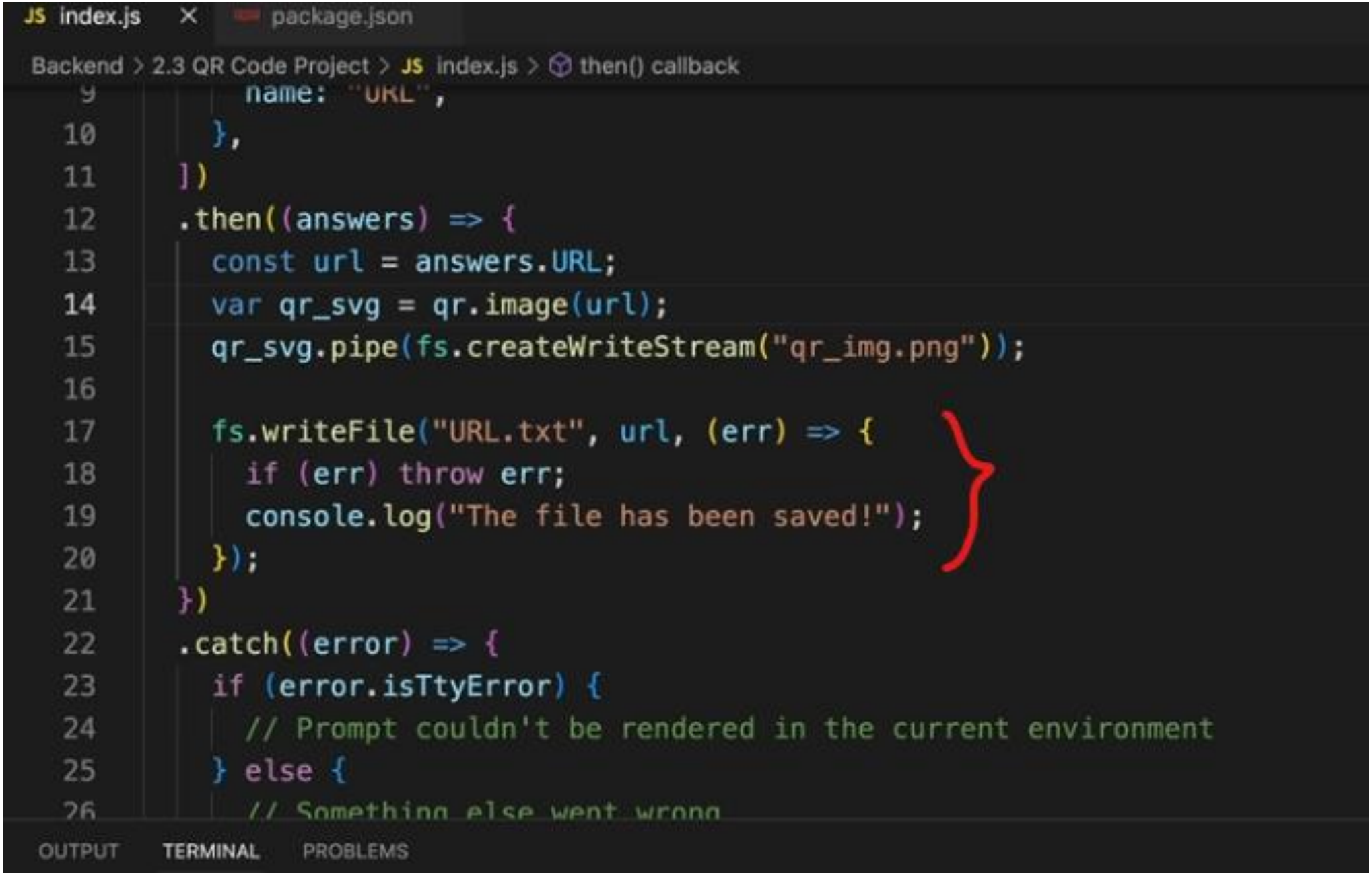
ndex.js × JS solution.js package.json

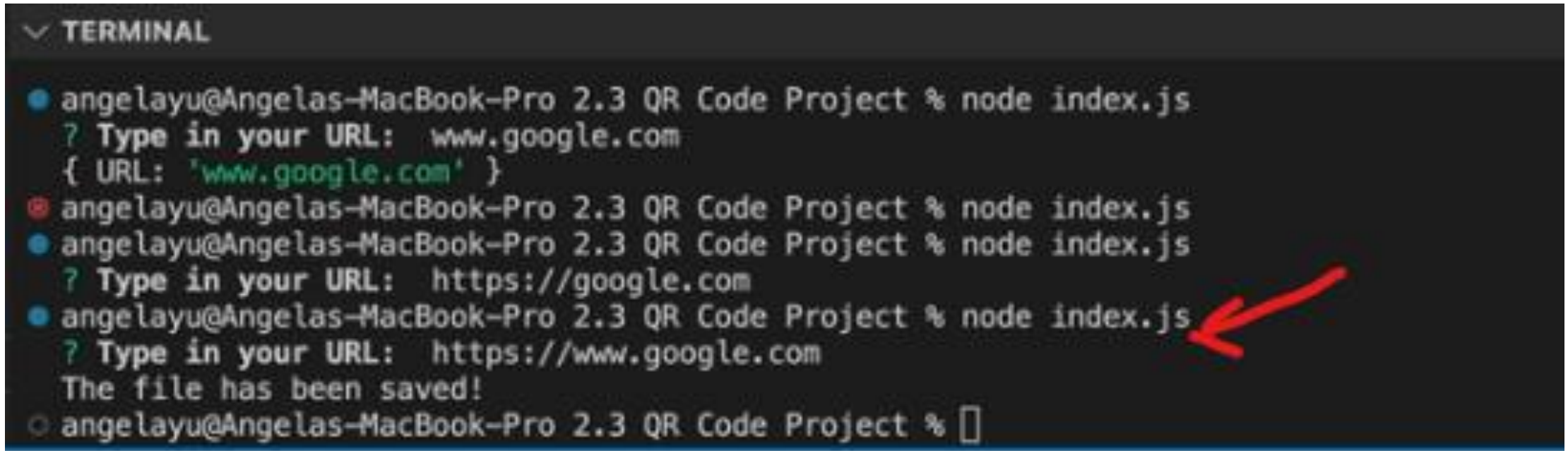
kend > 2.3 QR Code Project > JS index.js > ...

```
1  import inquirer from "inquirer";
2  import qr from "qr-image";
3  import fs from "fs";
4
5  inquirer
6    .prompt([
7      {
8        message: "Type in your URL: ",
9        name: "URL",
10     },
11   ])
12   .then((answers) => {
13     const url = answers.URL;
14     var qr_svg = qr.image(url);
15     qr_svg.pipe(fs.createWriteStream("qr_img.png"));
16   })
17   .catch((error) => {
18     if (error.isTtyError) {
19       // Prompt couldn't be rendered in the current environment
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```


EXPLORER
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package.json
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10.3 Mondrian Pr...
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11.1 Bootstrap La...
11.2 Bootstrap Co...
11.3 TinDog Project
12 Capstone 2 - ...
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ansi-regex
ansi-styles
base64-js
bl
buffer
chalk
chardet
cli-cursor
OUTPUT
TERMINAL
PROBLEMS
TERMINAL
angelayu@Angelas-MacBook-Pro Web Development Projects % cd '/Users/angelayu/Web Development Projects/Backend/2.3 QR Code Project'
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project % node index.js
? Type in your URL: www.google.com
{ URL: 'www.google.com' }
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project % node index.js
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project % node index.js
? Type in your URL: https://google.com


inquirer - npm
qr-image - npm
File system | Node.js v20.0.0
nodejs.org/api/fs.html#fswritefilefile-data-options-callback
Node.js
About this documentation
Usage and example
Assertion testing
Asynchronous context tracking
Async hooks
Buffer
C++ addons
C/C++ addons with Node-API
C++ embedder API
Child processes
Cluster
Command-line options
Console
Corepack
Crypto
Debugger
Deprecated APIs
Diagnostics Channel
DNS
Domain
Node.js v20.0.0 | ▸ Table of contents | ▸ Index | ▸ Other versions | ▸ Options
fs.writeFile(file, data[, options], callback)
▸ History
file <string> | <Buffer> | <URL> | <integer> filename or file descriptor
data <string> | <Buffer> | <TypedArray> | <DataView>
options <Object> | <string>
encoding <string> | <null> Default: 'utf8'
mode <integer> Default: 0o666
flag <string> See support of file system flags. Default: 'w'.
signal <AbortSignal> allows aborting an in-progress writeFile
callback <Function>
err <Error> | <AggregateError>
When file is a filename, asynchronously writes data to the file, replacing the file if it already exists. data can be a string or a buffer.
When file is a file descriptor, the behavior is similar to calling fs.write() directly (which is recommended). See the notes below on using a file descriptor.
The encoding option is ignored if data is a buffer.
The mode option only affects the newly created file. See fs.open() for more details.
import { writeFile } from 'node:fs';
import { Buffer } from 'node:buffer';
const data = new Uint8Array(Buffer.from('Hello Node.js'));
writeFile('message.txt', data, (err) => {
if (err) throw err;
console.log('The file has been saved!');
});


JS index.js
package.json
Backend > 2.3 QR Code Project > JS index.js > then() callback
name: "URL",
},
])
.then((answers) => {
const url = answers.URL;
var qr_svg = qr.image(url);
qr_svg.pipe(fs.createWriteStream("qr_img.png"));
fs.writeFile("URL.txt", url, (err) => {
if (err) throw err;
console.log("The file has been saved!");
});
})
.catch((error) => {
if (error.isTtyError) {
// Prompt couldn't be rendered in the current environment
} else {
OUTPUT
TERMINAL
PROBLEMS


TERMINAL
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project % node index.js
? Type in your URL: www.google.com
{ URL: 'www.google.com' }
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project % node index.js
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project % node index.js
? Type in your URL: https://google.com
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project % node index.js
? Type in your URL: https://www.google.com
The file has been saved!
angelayu@Angelas-MacBook-Pro 2.3 QR Code Project %